

ORACLE 2019 OLD PAPER

Q1. Attempt the following:

(1) Full form of SQL.

ANS:- SQL- Structured query language.

(2) Full form of DBMS.

ANS:- DBMS-Database management system.

(3) ER stands for -----

ANS:- Entity relationship.

(4) Define term database.

ANS:- Collection of meaningful data is called database.

(5) Define term primary key.

ANS:- A primary key is used to ensure data in the specific column is unique.

(6) ----- is symbol for concatenation operator in sql.

ANS:- ||

(7) -----operator is used for search specific pattern from the table.

ANS:- Like

(8) Define term synonyms.

ANS:- Synonyms is an alias or an alternate name for a table, view, sequence, etc.

(9) Define Floor().

ANS:-The floor() function returns rounded nearest small value of given number.

(10) Define ceil().

ANS:- The ceil() function returns rounded nearest big value of given number.

(11) ---- command is used to display the structure of a Table.

ANS:- describe (desc)

(12) Full form of DCL.

ANS:- DCL:-Data control language.

(13) Full form of DML.

ANS:- DML-Data manipulation language.

(14) Full form of PL/SQL.

ANS:- PL/SQL-Procedural language/Structured query language

(15) Define term Index.

ANS:- An index is a object that contains an entry for each value that appears in the indexed columns of the table and provides direct, fast access to rows.

(16) Define term commit.

ANS:- Commit is a transaction control language in sql. That permanently save all the changes made in the transaction of a database or table.

(17) Inner join also called----

ANS:- Equi join

(18) ---- command is used to display record from the table.

ANS:-Select

(19)--- command is used to edit record .

ANS:- Update

(20) DDL stands for -----

ANS:- Data definition language.

Q2 (A) Attempt any three.

(1) Explain select command.

ANS:-The SELECT command is a SQL command that is used to retrieves command.

The syntax for a SELECT command is as follows :-

SELECT column_list FROM table_name

[WHERE condition]

[ORDER BY column-name].

(2) Explain insert command.

ANS:- The INSERT command is data manipulation commands, which is used to manipulate data by inserting the information into the tables.

This command is used to add records to a table.

The syntax for a INSERT command is as follow:-

INSERT into table_name(column1,column2)

VALUES (value1, value2);

(3) Explain ALTER command.

ANS:- The ALTER command is used to add, delete or modify columns in an existing table. The ALTER command is also used to add and drop various constraints on an existing table.

The syntax for a ALTER command is as follows :-

ALTER TABLE table_name

ADD column_name datatype;

(4) Explain DELETE command.

ANS:- The DELETE command is a data manipulation command which is used to remove from a table.

The syntax for a DELETE command is as follow:-

DELETE FROM table_name

WHERE condition;

(5) Explain UPDATE command.

ANS:- The UPDATE command is a data manipulation command which is used to edit the records of a table.

It is used along with the SET clause, operationally, a WHERE clause may be used to match condition.

The syntax for a UPDATE command is as follow:-

UPDATE table_name

SET column1 = new_value1

WHERE condition;

(6) Explain Group By and Having.

ANS:- The GROUP BY statement groups rows that have the same values into summary rows.

The GROUP BY statement is often used with aggregate functions (COUNT,

MAX, MIN, SUM, AVG) to group the result-set by one or more columns.

The syntax for Group By :-

SELECT column_name

FROM table_name

GROUP BY column_name

ORDER BY column_name;

HAVING CLAUSE :-

Having clause special use for give condition to group by like WHERE.

You use the HAVING clause to filter groups of row GROUP BY can be used without HAVING. But HAVING must be used with GROUP BY.

You place the HAVING clause after your GROUP BY clause.

HAVING Syntax

SELECT column_name

FROM table_name

GROUP BY column_name

HAVING condition

ORDER BY column_name;

Q3. Attempt any three.

(1) Explain any two constrain in sql.

ANS:- constrain are set of rules used enforce data integrity.

- They don't take space in database like data.
- Even if single constrain ,entire record be rejected by oracle.

[1]Primary key:

- It can allow for single column.
- It dose not allow duplicate values.
- Null values are not allowed.
- It can identify the unique record in the table.

Example:

Create table stud

(sno number(3) primary key,

Sname varchar2(20),

City varchar2(10));

[2] foreign key:

- Foreign key is a field in a table which uniquely identifies each row of a another table.

- It is also known as reference key.
- The foreign field must be primary key .
- Child table may be duplicate and null values.

Example:

Create table stud

(s1 number(3),

S2 number(3),

Total number(6),

Sno number(6) references stud(sno));

(2) Explain data types in sql.

ANS:- A data type is a classification of a particular type of information or data.

- Each value manipulated by oracle has a data type.

[1]char:

Size is 1 to maximum 2000 characters and if data is short than its specified then space padded rest of size. it means char is used for fixed-length string.

[2]varchar/varchar2:

- Both varchar and varchar2 are same . it store alphanumeric data and also store data.
- In varchar data is short than its length it can not remove extra length whereas varchar2 remove extra length.
- Varchar2 used for variable-length strings.

[3]Long:

- Long datatype is large object (LOB) datatypes.
- It stored up to 2gb length.

- But restriction is you cannot use WHERE clause, group by clause, order by clause, distinct operator in select statement.

[4]Number:

- This datatype used to store negative and positive integers: fixed point or floating numbers with precision of up to 38 digits.
- A data type is a classification of a particular type of information or data.
- Each values manipulated by oracle has a data type.

(3) SQL v/s PL/SQ

SQL stands for structured query language.	PL/SQL stands for procedural language/structured query language.
Dose not have procedural capacity.	It has procedural capacity.
It is used to create application.	It is used to maintain data.
SQL is a subset of PL/SQL.	PL/SQL is a superset of SQL.
It is not a block structured language.	It is a block structured language.
SQL is a command interpreter.	PL/SQL is not a command interpreter.

(4) What is package?

ANS:- A package ia set of pl/sql statement .

- It is an object which can hold other objects like procedure, function, cursor, exception and so on.
- It is a container object and allow for all the object to bind together.

➤ Components:

[1]package specification:

- It is known as package header.
- It can contain the name of package.
- It doesn't contain any code for procedure and function.
- Object must be declare in this section before it is used.
- Syntax:
 - Create or replace package package_name
 - [IS/AS]
 - PL/SQL statement;
 - End package name;

[2] package body:

- It can contain the definition of public object that are declare in the specification.
- It can also have other object, which are private to the package.
 - Syntax:
 - Create or replace package body
 - Package_name [is/as]
 - PL/SQL statement;
 - End package;
- To call the package:
 - The package object can be executed by two keywords like exec or call.
 - Eg:
 - Exec jpack.sum;
- Advantages:
 - 1] Package is more efficient because it is modual programming.
 - 2] The scope of package is local as well as global.
 - 3] Easy to understand.

(5) Explain union clause.

ANS:- The union operator is used to combine the results of two or more select queries into a single result set.

- It can Combines two or more queries and removes the duplication.
- The union operator creates a new table by placing all rows from two source tables into a single result table.

➤ Rules:

- 1] The number and the order of the columns must be the same in all queries.
- 2] The columns must also have similar data types.
- Syntax:
 - Select column_list from table_name
 - Union
 - Select column_list from table_name;
- Example:
 - Take two tables:

- Employee

=====		
Emp_id	name	dept_id

1	Abc	1
2	Urm	2
3	Jfjg	2
4	Jjy	1
5	Iek	3
6	Ktk	5

- Department

=====	
Dept_id	dept_name

1	it
2	hr
3	medical
4	quality

- select dept_id from employee
union
select dept_id from department;
- Output:

- 1
- 2
- 3
- 4
- 5

(6) Explain sub-query.

ANS:- A subquery is a SQL query nested inside a larger query.

```

SELECT  select_list
FROM    table
WHERE   expr operator
        (SELECT  select_list
         FROM    table);

```

☐ Syntax :

☐ The subquery (inner query) executes once before the main query (outer query) executes.

☐ The main query (outer query) use the subquery result.

☐ A sub-query answers the queries that have multiple parts

☐ The sub-query answers one part of the question and the parent query answers the other part.

☐ **Types of sub-query:**

☐ Single row sub queries

☐ Multiple value sub queries

☐ Multiple column sub queries

☐ Multiple row sub queries

[1] Single row sub queries:

☐ It returns single value from a table.

☐ Example

☐ Select ename, sal from emp

☐ Where sal = (select max(sal) from emp);

☐ Output

☐ ENAME SAL
☐ -----
☐ KING 5000

[2] Multiple value sub queries:

☐ This returns more than one value to the parent query statement.

☐ IN operator is the most commonly used in multiple row sub query

☐ Example

- Select ename, sal, job from emp
- Where sal in(select max(sal) from emp group by job);

☐ **Out put**

ENAME	SAL	JOB
MILLER	1300	CLERK
ALLEN	1600	SALESMAN
KING	5000	PRESIDENT
JONES	2975	MANAGER
FORD	3000	ANALYST
SCOTT	3000	ANALYST

[3] Multiple column sub queries:

☐ This returns values more than one column in subquery

☐ Example

- Select ename, sal , job from emp
- Where (sal,job) in (select max(sal),job from emp group by job);

☐ **Output :**

ENAME	SAL	JOB
MILLER	1300	CLERK
ALLEN	1600	SALESMAN
KING	5000	PRESIDENT
JONES	2975	MANAGER
FORD	3000	ANALYST
SCOTT	3000	ANALYST

[4] Multiple row sub queries:

- This type of sub-query returns more than one row of result from the sub-query

- Example

- Select ename, deptno from emp
- Where deptno not in (select deptno from emp where ename='SCOTT');

- **Output**

ENAME	DEPTNO
ALLEN	30
WARD	30
MARTIN	30
BLAKE	30
CLARK	10
KING	10
TURNER	30
JAMES	30
MILLER	10

(C) Attempt any two

(1) Explain Dr.E.F Codd rules.

Ans:- •Rule 1-Information Rule:-

All data should be presented in table.

•Rule 2-Guaranteed Access Rule:-

All data should be accessed without any problem.

•Rule 3-Systematic Treatment of NULL values:-

A field should be empty(null) which is different from 0(zero) or empty field, this can not apply to primary key.

•Rule 4-Dynamic online catlog on the basis of relational model:- A relational database must provide access to its structure through the same tools that are used to access the data.

- Rule 5-Comprehensive data sublanguage rule:-

The database must be support at list one clearly defined language that includes functionality for data definition or manipulation or integrity.

- Rule 6-View updating rule:-

Each view should be support the same full range of data manipulation that has direct access to a table available.

- Rule 7-High-level insert, update, delete:-

This rule states that insert, update, delete should be supported for any retrievable rather than just for a single row in a single table.

- Rule 8-Physical data independence rule:-

The user should remain isolated from the physical method of sorting and retrieving information from the database.

- Rule 9-Logical data independence:-

How data is accessed should not be changed when the logical structure of database changes.

- Rule 10-Integrity independence:-

The database language like SQL should support all types of user condition.

- Rule 11-Distribution independence:-

It is possible that the data can be distributed easily.

- Rule 12-Non-subversion rule:-

There should be no way to modify the database structure other than the multiple row database language like SQL.

(2) Explain IN, ANY, BETWEEN in SQL.

ANS:-

IN:-

- IN is an operator in SQL, which is generally used with a where clause.
- Using IN operator, multiple values can be specified.
- It allows us to easily test if an expression matches any value in a list of values.
- IN operator used to replace many or conditions.

Syntax

Select column_name from table_name

Where column_name in (value1, value2....);

Any:-

- The ANY operator compares the given value to each subquery value and returns those values that satisfy the condition.
- It always evaluates to TRUE if at least one sub query value matches according to the given condition.

Syntax

Select column_name from table_name

Where column_name any (select column_name from table_name where[condition]);

ALL:-

- ALL operator is used with select, where and having statements.
- ALL means that the condition will be true only if the operation is true for all values in the range.

Syntax

Select column_name from table_name

Where column_name all (select column_name from table_name
Where[condition]);

BETWEEN:-

- The BETWEEN operator selects values within a given range. The values can be numbers, text, or dates.
- The BETWEEN operator is inclusive: begin and end values are included.

Syntax

Select column_name from table_name

Where column_name between value1 and value2;

(3) Explain Index.

Ans:- An index is an object that contains an entry for each value that appears in the indexed column of the table and provides direct, fast access to rows.

Types of index:- a)B-tree index

- B-tree is a self-balancing tree data structure that keeps data stored and allows searches, sequential access, insertion, and deletions in time.
- Each page in an index B-tree is called an index node. The top node of the B-tree is called the root node. The bottom nodes in the index are called the leaf nodes.

b)Bitmap index

- A bitmap is a mapping from one system such as integers to bits. It is also known as bitmap

index or a bit array.

- The memory is divided into units for bitmap.

- These units may range from a few bytes to several kilobytes.
- The bitmap provides a relatively easy way to keep track of memory as the size of the bitmap is only depend on the size of the memory and the size of units.

c) Functional based index

- Function-based indexes allows you to create an index based on a function or expression.
- Function-based indexes can involve multiple columns, arithmetic expression, or may be a pl/sql function.

Syntax

Create index column_name on table_name(upper(column_name));

d) Application domain indexes

- Application domain indexes are custom indexes that are specific to the needs of your application.
- They can be based on the semantics of your data and queries.

(3) Explain commit, rollback, savepoint.

Ans:- Commit:- Commit is a transaction control language in SQL. It lets a user permanently save all the changes made in the transaction of a database or table. Once you execute the commit, the database cannot go back to its previous state in any case.

Syntax

Commit;

Rollback:-

Rollback is a transactional control language in SQL. It lets a user undo those transactions that aren't saved yet in the database. One can make use of this command if they wish to undo Any changes or alterations since the execution of the last commit. Syntax

rollback;

Savepoint:-

A savepoint is a point in a transaction when you can roll the transaction back to a certain point without rolling back the entire transaction.

Syntax

Savepoint savepoint_name;

This command servers only in the creation of a savepoint among all the transactional

statements. The rollback command is used to undo a group of transactions. Rollback to savepoint_name;

(5) Explain PL/SQL block structure in detail.

ANS:- In PL/SQL, all statements are classified into units that is called blocks. PL/SQL blocks can include variables, conditional statements and exception handling. Blocks can also build a function or a procedure or a package.

The declaration section:-

Code block start with a declaration section, in which memory variables, constants, cursors and other oracle objects can be declared and if required initialize.

The begin section:-

Consist of set of SQL and PL/SQL statements, which describe processes that have to be applied to table data. Actual data manipulation, retrieval, looping and branching constructs are specified in this section.

The exception section:-

The section deals with handling errors that arise during execution data manipulation statements, which make up PL/SQL code block. Errors can arise due to syntax, logic or validation rule.

The end section:-

This marks the end of PL/SQL block.

Example:- declare

y number(5);

bcheck exception;

begin

y:=&y;

If y<18 then

raise bcheck;

else

dbms_output.put_line('you can vote');

end if;

exception

when bcheck then

dbms_output.put_line('you cannot vote');

end;

3 (A) Attempt any three

(Q1) Explain Import and Export.

ANS:- Exporting data is the process of converting data from a database into one of many interchange data formats so that the data can be used by systems that do not interface directly with the database or to be imported in to another database.

Importing data is the process of taking data presented in some interchange format and converting it for use within a database system.

(Q2) What is view with example?

ANS:- A view is a virtual table that stores only definition in the form of SQL statement. It is used to hide information from other user.
Syntax

Create [or replace][force] view < view _name> as

Select column _name from table _name Where[condition];

Example

Create table student

(Sno number(3), Sname varchar2(20), Salary number(5));

Create or replace view v1 as

Select salary from student Where salary>1000

With read only;

(Q3) Explain lock in concurrency control.

ANS:- A Lock is a variable assigned to any data item in order to keep track of the status of that data item so that isolation and non-interference is ensured during concurrent transactions.

(Q4) Explain Grant command.

ANS:- SQL Grant command is specifically used to provide privileges to database objects for a user. This command also allows users to grant permissions to other users too.

Syntax:

grant privilege_name on object_name to {user_name | public | role_name}

Here privilege_name is which permission has to be granted, object_name is the name of the database object, user_name is the

user to which access should be provided, the public is used to permit access to all the users.

(Q5) Explain Revoke command.

ANS:- Revoke command withdraw user privileges on database objects if any granted. It does operations opposite to the Grant command. When a privilege is revoked from a particular user U, then the privileges granted to all other users by user U will be revoked.

Syntax:

```
revoke privilege_name on object_name from {user_name | public |  
role_name}
```

(Q6) What is redo logs files?

ANS:- •Redo logs files are operating system files used by oracle to maintain logs of all transactions performed against the database.

•The primary purpose of these log files is to allow oracle to recover changes made to the database in the case of a failure. •During crash recovery, the redo log is crucial for replaying the logged changes to bring the database to a consistent state.

•Redo log files are created when the database is created.

3(b) Attempt any three

(1) Explain SQL *loader.

ANS: SQL loader utility is used to load data from other data source into oracle.

* SQL loader provides several methods to load data.

*SQL loader is uses in two ways with or without control file.

*Features of SQL loaders:

- 1.Load data from multiple data files during the same load session.
- 2.load data into multiple tables during the same load session.
- 3.Specify the character set of data.

4.Manipulate the data before loading it , using SQL functions.

5.Load data from data.

6.Selectively load data.

(2) Differentiate: function v/s procedure

ANS:

Function	Procedure
Functions take parameters and return a value.	Procedure does not take any parameters and does not return value.
Function does not allows DML statements.	Procedure allows DML statements.
A function has a return type.	A procedure does not have return type.
Functions can be called from procedure.	Procedure cannot be called from a function.
We cannot use transaction in function.	We can use transaction in procedure.

(3) Explain varray.

ANS:-Varray stand for "variable size array".

* The varray is one of the three types of collection in pl/sql.

*The Varray's key feature is that when you declare varray type,you specify the maximum number of elements that can be defined in the varray.

* A varray type is created with the create type statement.

* A varray is single-dimensional collectios of elements with the same data type.

* Varray is used more frequently when the size of the array is known to us.

* You must specify the maximum size and the type of elements.

*Syntax:-

Create or replace type type_name is varray (n) of data_type;

* 'n' is the number of elements (maximum) in varray.

(4) Explain sequence with appropriate example.

ANS: A sequence is a set of integers that are generated in order on demand.

- * Sequence are frequently used in many databases.

- * All sequence attributes are optional and they all have default values.

- * Each sequence has two columns: NEXTVAL and CURRVAL.

- * NEXTVAL is used for next value while CURRVAL is used for current value of data in sequence.

- * Sequence can be created, changed and dropped with the following three SQL commands.

- 1 Create Sequence

- 2 Alter Sequence

- 3 Drop Sequence

- * Syntax:- Create sequence sequence_name

Start with <no>

Increment by <no>

Minvalue <no>

Maxvalue <no>

Cycle

Cache <no>;

- * Example:

Create Sequence s1

Start with 5

Increment by 5

Minvalue 1

Maxvalue 20

Cycle

Cache 2;

Select s1.nextval from dual;

(5) Explain %type and %rowtype

ANS: %TYPE:-

- * %type is used to declare the variable of the same data type as the database column.

- * This is called dynamic data type assignment.

- * Changes in table data type direct reflected in the program.

- * Syntax:-Variable table_name.field_name%type;

%ROWTYPE:-

- * %rowtype is used to fetch the entire rows of the table.

- * With the help of this,we don't have to describe each field in the record definition.

- * It means oracle select data type automatically.

- * Syntax:-

Variable table_name%rowtype;

(6) Explain exception handling in PL/SQL. ANS:-

- * An exception is an error which disrupts the normal flow of program instructions.

- * PL/SQL provides exception block which raises the exception thus helping to find out the fault and resolve it.

* There are two type of exceptions:-

1 System Defined Exception

2 User Defined Exception

* No_data_found:

* It is a user defined exception.

* Which is generally use for select statements,when no record match we can use this exception.

* To create user defined exception just follow this steps:

1 Declare exception in declaration part.

2 Raise declared exception in execution part.

3 In last exception part put your declared exception in between "when" and "then" for proper message.

3(c) Attempt any three :

(1) Explain package with example.

ANS:- A package is an object, which holds other objects.Like procedure, functions, cursor, exception etc.

-> Components of package

☐ Package specification

☐ Package body

--> package specification :-

☐ Contains name of the package.

☐ Contains declaration of procedure, function, cursors, exception etc.

☐ It does not contain any code for procedures/functions.

--> syntax of specification of package:-

create [or replace] package <package name> {as/is}

<statements>

End [package name];

--> package body :-

- Package body contains the definition of public objects that are declared in the specification.

- Private objects are defined in package body only and they are not visible/accessible outside the package.

- The variables declared in the package specification can be accessed by any procedure/function within the package.

- Package object referred as

- exec <package name>.<object name>;

--> package body syntax:-

create [or replace] package body <package name> {as/is}

<statements>

End [package name];

--> Example:-

-- create package:-

Create or replace package jpack as

procedure jemp(jno emp.empno%type);

end jpack;

-- create package body:-

Create or replace package body jpack as

Procedure jemp(jno emp.empno%type) as

Jname emp.ename%type;

Begin

Select ename into jname from emp where empno=jno;

dbms_output.put_line('Employee '||jname||' found');

End;

End jpack;

/

-- Execute:-

exec jpack.jemp(7787);

(2) Explain triggers with example.

ANS:- Triggers are the programs that are executed automatically in response to a Change in database.

☐ Oracle allows special types of procedures that are automatically executed when events like insert/update/delete occurs. These event procedures called 'Database Triggers'.

☐ You may not use commit, rollback and savepoint statement within trigger blocks.

☐ : old and :new are concept of Trigger.

☐ The :new refers to a record containing the new, post change data.

☐ The :old refers to a record containing the old, pre-change data.

--> Types of Triggers:-

☐ Before update statement level

☐ After insert row level

☐ Before delete row level

--> Trigger predicates :-

☐ Inserting :- True if the trigger statement is Insert

- ❑ Deleting :- True if the trigger statement is Delete
- ❑ Updating :- True if the trigger statement is update

--> syntax of Trigger :-

create [or replace] trigger <trigger name>

[Before/after/instead of]

[Delete/insert/update [of <col1>, <col2>...]]

On <table/view name>

[For each row [when <condition>]]

Declare

<Variable>

Begin

<pl/sql statement>

Exception

<exception>

End;

--> Example :-

create or replace trigger jmsg

before insert or delete or update

On dual

For each row

Begin

If inserting then

```
dbms_output.put_line('record inserted in dual');
```

Elself deleting then

```
dbms_output.put_line('record deleted in dual');
```

Elself updating then

```
dbms_output.put_line('record updated in dual');
```

```
End if;
```

```
End;
```

(3) Explain backup & recovery in Oracle.

ANS:-

Backup :-

- A backup is a copy of data. This copy can include important part of database such as the control file and data files.

- A backup is a safeguard against unexpected data loss and application errors.

- If you loss the original data, then you can reconstruct it by using a backup.

--> Types of backup :-

- Backup of oracle data are either physical or logical.

1. Physical oracle backups :-

- Cold backup

- Hot backup

- Control file backup

- Redo log file backup

2. Logical backup :-

- Using export/ import

--> Features of backup :-

- It makes the process of data recovery simple and easy.
- It provides data security to the users.
- It is a cost-effective process to retrieve the data.

Recovery :-

- Recovery is use to store a backup of control file and reconstruct it and make it available to the oracle database server.

--> Features of recovery :-

- The process of recovering is expensive.
- When there is failure, it refers to recovering the lost data.
- It increases the database reliability.

(4) Explain cursor.

ANS:- when you execute a SQL statement from PL/SQL, the oracle assigns a private work area for that SQL statement is known as a cursor.

- The pl/SQL is mechanism by which you can name that work area and manipulate the information within it.
- Data stored in the cursor known as Active Data Set (ADS).

--> Types of cursor :-

1. Implicit cursor :- The implicit cursor are automatically generated by oracle while an SQL statement is executed.
2. Explicit cursor :- A cursor can also be opened for processing data through pl/SQL block such as user-defined cursor known as explicit cursor.

--> Attributes of cursor :-

- %isopen - returns true if cursor open, otherwise false.

□ %found - returns true if record was fetched successfully, otherwise false.

□ %notfound - returns true if record was not fetched successfully, otherwise false.

□ %rowcount - returns number of records processed from the cursor.

--> Steps :-

1. Declaring the cursor :-

ex :-

```
cursor c_customers is select id, name from customers;
```

2. Opening the cursor :-

ex :-

```
open c_customers;
```

3. Fetching the cursor :-

ex :-

```
fetch c_customers into c_id, c_name;
```

4. Closing the cursor :-

ex :-

```
close c_customers;
```

--> Example :-

Declare

```
j_nam emp.ename%type;
```

```
j_sal emp.sal%type;
```

```
Cursor j_emp is select ename, sal from emp;
```

```
begin
```

```
open j_emp;
```

```
fetch j_emp into j_nam, j_sal;  
dbms_output.put_line('salary of '||j_nam||' is '||j_sal);  
close j_emp;  
end;
```

(5) Explain Instance Architecture.

ANS:- when a database is started on a database server, Oracle allocates a memory area called the System Global Area (SGA) and starts one or more oracle processes.

- This combination of SGA and Oracle processes is called an oracle Instance.

--> The Instance and The Database :-

- After starting an instance, oracle associates the instance with the specified database. This is called mounting the database.

- The database then ready to be opened, which makes it accessible to authorized users.

- Only the database administrator can start up an instance and open the database.

- If a database is open, then the database administrator can shutdown is controlled through connection to oracle with administrator previlleges.

- Normal users do not have control over the current status of an oracle database.

--> An oracle Instance is build up by considering following aspects :-

1. Database process

2. Memory structure

3. Data files

1. Database process :-

□ The oracle RDBMS uses two types of processes, user process and oracle process (called background process).

1. User process

2. Oracle process

□ Server process

□ Background process

2. Memory structure :-

□ Oracle uses memory to store information such as :

□ Program code.

□ Information about connected session.

□ Information needed during program execution.

3. Data files :-

□ The oracle architecture are the physical files where our information stored on disk.

□ Oracle has several data files :-

1. Database data files

2. Control files

3. Online redo log files

4. Other database related files.

RDMS Using ORACLE-2019 New

Q1-A

- (1). Row types are used to store which type of data ?
Ans. Multiple data values
- (2). A table can't have one more than one long data type (True/False).
Ans. false .
- (3). _____ removes all the rows from table, but its structure and constraints and so on, remains.
Ans. TRUNCATE TABLE statement
- (4). In SQL, which command is used to select only one copy of each set of duplicate rows ?
Ans. DISTINCT COMMAND

Q1-B

1. Write a note on E-R Diagram.

- E-R Diagram stands for Entity Relationship Diagram.
- E-R Diagram is a diagram that represents relationships among entities in a data
- **Rectangles** represent table sets.
- **Diamonds** represent relationship between tables.
- **Lines** link attributes represent connection between tables with relationship and field.
- **Ellipses** represent table fields
- **Underline** indicates primary key attributes

2. Differentiate: DBMS V/S RDBMS.

DBMS	RDBMS
RELATIONSHIP BETWEEN TWO TABLES OR FILES ARE MAINTAINED PROGRAMICALLY	RELATIONSHIP BETWEEN TWO TABLES OR FILE CAN BE SPECIFIED AT THE TIME OF TABLE CREATION
DOES NOT SUPPORT CLIENT /SERVER ARCHITECTURE	SUPPORTS CLIENT/SERVER ARCHITECTURE
DOES NOT SUPPORT DISTRIBUTED DATABASES	SUPPORTS DISTRIBUTED DATABASES
NO SECURITY OF DATA	MULTIPLE SECURITY LEVES; 1 LOGIN AT OS LEVEL 2 COMMAND LEVEL SECURITY 3 OBJECT LEVEL SECURITY
EACH TABLE IS GIVEN EXTENSION	MANY TABLES ARE GROUPED IN ONE DATABASE IN RDBMS

Q1-C

1.Explain various components of sql.

- There are four types of SQL command.**

DDL (Data Definition Language)

DML (Data Manipulation Language)

DCL (Data Control Language)

DQL (Data Query Language)

- DDL (Data Definition Language)**

- CREATE:** It is used to create objects (table, view, function, procedure, trigger etc.) in the database.

- ALTER:** It is used to change the structure of database.

- DROP:** It is used to delete objects from the database.

- TRUNCATE:** It is used to remove all records from the table, including all spaces allocated for the records are removed.

- GRANT:** It is used to give different command grant (access rights) to the user.

- REVOKE:** It is used to take the given grant back from the user.

- DML (Data Manipulation Language)**

- INSERT:** it used to insert the data into a table

- UPDATE:** it used to update the existing data within a table.

- DELETE:** it used to delete all the records from the table, the space will be remaining.

- CALL:** it is used to call a PL/SQL program.

- DCL (Data Control Language)**

- COMMIT:** it is used to save the work.

- SAVEPOINT:** it is used to identify a point in transaction.

- ROLLBACK:** Restore database to original since the last commit (like undo).

- DQL (Data Query Language)**

- SELECT:** It is used to retrieve data from the database.

2. What are E-F code rules? Explain.

- Rule 1: The Information Rule – All data should be presented in table form.

- Rule 2: Guaranteed Access Rule – All data should be accessible without ambiguity. This can be accomplished through a combination of the table name, primary key and column name.

- Rule 3: Systematic Treatment of Null Values –A field should be allowed to remain empty. This involves the support of a null value, which is distinct from an empty string or a number with a value of zero.

- Rule 4: Dynamic On-Line Catalog based on the Relational Model –A relational database must provide access to its structure through the same tools that are used to access the data. This is usually accomplished by storing the structure definition within special system tables.
- Rule 5: Comprehensive Data Sublanguage Rule –The database must support at least one clearly defined language that includes functionality for data definition, data manipulation, data integrity, and database transaction control.
- Rule 6: View Updating Rule –Data can be presented in different logical combinations called views. Each view should support the same full range of data manipulation that has direct access to a table available. In practice, providing update and delete access to logical views is difficult and is not fully supported by any current database.

Q1-D

1. List out various operators and explain any 2 operators with example.

- An operator is a symbol that tells the compiler to perform specific mathematical or logical manipulation.
- PL/SQL language is rich in built-in operators and provides the following types of operators
- **Comparison operators**
- **Arithmetic operators**
- **Relational operators**
- **Logical operators**
- **Types Of Logical operators:**

Operator	Description	Examples
• and	• Called the logical AND operator. If both the operands are true then condition becomes true.	• (A and B) is false.
• or	• Called the logical OR Operator. If any of the two operands is true then condition becomes true.	• (A or B) is true.
• not	• Called the logical NOT Operator. Used to reverse the logical state of its operand. If a condition is true then Logical NOT operator will make it false.	• not (A and B) is true.

- Types of Comparison operators
 - IN
 - Between
 - Like
- The IN operator is used when you want to compare a column with more than one value. It is similar to an OR condition.
- Example
 - select *From emp
 - where deptno in(10,20);
- The operator BETWEEN and AND, are used to compare data for a range of values.
- This operator also known as range operator.
- Example
 - select *From emp
 - where sal between 800 and 1500;
- This operator is known as Pattern Matching operator
- The LIKE operator is used to list all rows in a table whose column values match a specified pattern.
- Example
 - SELECT *FROM EMP
 - WHERE ENAME LIKE 'S%';

3. Explain various data type available in SQL.

- **char**
 - size 1 to maximum 2000 characters and if data is short than its specified than space padded rest of size
- **Varchar**
- **Varchar2**
 - Both varchar and varchar2 are same, it stores alphanumeric data and also store data in its size if data is short than is length than varchar2 remove extra length of data type and allocates to other data

- **Long**
- Long data type is Large Object (LOB) data types. It stored up to 2GB length. But restriction of this type is you cannot use WHERE clause, Group By clause, Order By clause, Distinct operator in SELECT statement
- **Number**
- This data type used to store negative and positive integers: fixed-point and floating numbers with precisions of up to 38 digits.
- **Date**
- The date data type is used to store date and time information. Default date format is DD-MON-YY.

Q2-A

- 1. Clause acts like a where clause but is used for groups rather than rows.
Ans.HAVING
- 2.The condition in a where clause can refer to only one value. (True / False)
Ans.False
- 3.Drop table cannot be used to drop a table referenced by a constraints.
Ans.Foreign key
- 4.The statement in SQL which allows changing the definition of a table is
is .

Ans.Alter table

Q2-B

1. Explain group by and having clause with suitable example.

- The group by statement groups rows that have the same values into summary rows, like “find the maxsal from each job in emp table”.
- The group by statement is often used with aggregate functions (count, max, sum, avg) to group the result-set by one more columns.
- By default, group by sorts the rows in ascending order based on the values in the group column

- Group by syntax:

```
select <column_name> from <table_name>
group by <column_name> order by <column_name>;
```
- Having clause special use for give condition to group by like where.
- You place the having clause after your group by clause.
- Having syntax:

```
select <column_name> from <table_name> group
by <column_name> having <condition> order by <column_name>;
```
- Example:

```
Select job, count(sal) group by job having='clerk';
```

2. Write a note on set operators.

- The SQL set operation is used to combine the two or more SQL SELECT statements.
- **Types of set operation**
 - **Union**
 - **Union all**
 - **Intersect**
 - **Minus**
- **Union**
- The sql union operation is used to combine the result of two or more sql select queries.
- The union operation eliminates the duplicate rows from its result set.
- Syntax:

```
select <column_name> from <table1>
union
select <column_name> from <table2>;
```
- **Union all**
- Union all operation is equal to the union operation. It returns the without removing duplication and sorting the data.
- Syntax:

```
select <column_name> from <table1>
union all
select <column_name> from <table2>;
```

- **Intersect**

- It is used to combine two select statements. The intersect operation returns the common rows from both the select statements.

- Syntax: select <column_name> from <table1>

- intersect

- select <column_name> from <table2>;

- **Minus**

- It combines the result of two select statements. Minus operator is used to display the rows which are present in the first query but absent in the second query.

- Syntax:

- Select <column_name> from <table1>

- Minus

- Select <column_name> from <table2>;

Q2-C

1. What is join ? Explain its various types.

- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.

- **Different types of sql joins**

- Inner join
- Outer join
 - Left outer join
 - Right outer join
 - Full outer join
- Self join
- Cross join

- **Inner join:**

- SQL INNER JOIN keyword

- The inner join keyword select records that have matching values in both tables.

- This join is also known as equi join.

- **Outer join**
- SQL left join keyword:
 - The left join keyword returns all records from the left table (table1), and the matched records from the right table (table2).
- SQL right join keyword:
 - The right join keyword returns all records from the right table (table2), and the matched records from the left table (table1).
- SQL full outer join keyword:
 - The full outer join keyword return all records when there is a match in either left (table1) or right (table2) table records.
- **Self join**
 - A self join is a regular join, but the table is joined with itself.
- **Cross join**
 - In this join there is no condition given so records multiply by each other table
 - Like if one table have 3 records and other table have 4 and we apply cross join than result will be 12 rows display.

2. What are constraints ? Explain any two constraints with suitable example.

- Constraints are set of rules used to enforce data integrity.
- They don't take space in database like data.
- Even if single column fails to satisfy constrain, entire record will be rejected by Oracle.
- **Type of constraints are:-**
 - Primary key
 - Foreign key
 - Unique
 - Not null
 - Check

- **Unique**
- The purpose of UNIQUE constraint is to ensure that the data in the column is unique and it must not be repeated across the column.
- A table can have more than one unique key column.
- Unique constraint at column level:
- Syntax: <column name data type (size)> unique,
- Unique constraint at table level:
- Syntax: unique(columnname[,columnname,])
- **Check**
- Business rule validations can be applied to a column using check constraint.
- Check constraint must be specified as logical expression that evaluates either to true or false.
- Check constraint at column level:
- Syntax: <column name data type (size)> check (logical expression)
- Check constraint at table level:
- Syntax: check(logical expression)

Q2-D

1. Consider these following tables and only solve the query.

Tables:1. student: s_id (Primary key), sname, city

2. result: r_id (Primary key), mark1, mark2, s_id (foreign key)

- Query:-
- Write a SQL statement to make a list with r_id, s_id, student name and mark1 for those whose marks between 70 and 90.
- Find out maximum mark2 from result table.
- Ans.
- Create table student
(s_id number(3) primary key,
Sname varchar2 (30),
City varchar2 (20));

- Create table result
(r_id number(3) primary key,
Mark1 number(3),
Mark2 number(3),
S_id number foreign key (s_id) references student (s_id));
- Select s.sname, r.mark1 from student s, result r
where s_id = s_id, between 70 and 90;
Select max(mark2) from result;

2. Table : Employee : emp_no (Primary key), emp_name, city, mgr_no, job, sal, dept_no

- Query:-
- Display department wise average salary.
- Ans.
- Create table employee
(emp_no number(4) primary key,
Emp_name varchar2(30),
Mgr_no number(5),
Job varchar2(20),
Sal number(5),
Dept_no number(5));
Select dept_no, avg(sal) from employee group by dept_no;

Q3-A

- 1. A view is virtual table that can be accessed via SQL command.(true/false) Ans.True
- 2. In a _____ two database operations wait for each other to release a lock. Ans. deadlock
- 3.Oracle uses a method called _____ to implement concurrency control.
Ans. Multi-version concurrency control
- 4.Indexes may be created or dropped at any time.(True False)
Ans.True

Q3-B**1. Differentiate : pessimistic lock v/s optimistic lock.**

pessimistic lock	optimistic lock.
Pessimistic locking is when you take an exclusive lock so that no one else can start modifying the record.	Optimistic locking is when you check if the record was updated by someone else before you commit the transaction.
Pessimistic locking aims to avoid conflicts by using locking.	Optimistic Locking allows a conflict to occur, but it needs to detect it at write time.

2. Define the term : cluster , synonym.

- **Cluster :-**
- Clusters in SQL are used to store data that is from different tables in the same physical data blocks.
- They are used if records from those tables are frequently queried together.
- By storing same data blocks, the number of database block reads needed to full fill such queries decreases which improves performance.
- Each cluster stores tables data and maintains a clustered index to sort data.
- Columns within the cluster index are called clustered keys. These determine the physical placement of rows within the cluster.
- Cluster key is usually a foreign key of one table that references the primary key of another table in cluster.
- **Synonym :-**
- A synonym is an alternative name for objects such as tables, views, sequences, stored procedures, and other database objects.
- Synonyms is dependent object.

Q3-C

1. What is view ? Explain in detail.

- A view is a virtual table that stores only definition in the form of SQL statement. It is used to hide information from other user.
- **Feature of view : -**
- Creating view does not take any storage space in oracle
- The maximum number of columns that can be defined in a view is 1000
- When new records are inserted/updated/deleted in base table, the changes are automatically reflected in the view
- View once defined, can be used in Joins and Sub-query also
- **Types of view :-**
- Read only view
- Updateable view
- **Syntax :-**
- `CREATE [OR REPLACE] [FORCE] VIEW <View Name> AS
SELECT <Fields> FROM <Table> WHERE <condition>`

2. Write a note on indexes.

- An index is a object that contains an entry for each value that appears in the indexed column(s) of the table and provides direct, fast access to rows.
- **B tree indexes :-**
- B-tree is a self-balancing tree data structure that keeps data sorted and allows searches, sequential access, insertions, and deletions in time.
- Each page in an index B-tree is called an index node. The top node of the B-tree is called the root node. The bottom nodes in the index are called the leaf nodes.
- **Bitmap indexes :-**
- A bitmap is a mapping from one system such as integers to bits. It is also known as bitmap index or a bit array.
- The memory is divided into units for bitmap.
- These units may range from a few bytes to several kilobytes.
- Each memory unit is associated with a bit in the bitmap.

- If the unit is occupied, the bit is 1 and if it is empty, the bit is zero.
- The bitmap provides a relatively easy way to keep track of memory as the size of the bitmap is only dependent on the size of the memory and the size of the units.
- **Functional based index :-**
- Function-based indexes allow you to create an index based on a function or expression.
- Function-based indexes can involve multiple columns, arithmetic expressions, or maybe a PL/SQL function
- The following example shows how to create a function-based index:
Create index emp_idx on emp (upper(ename));
- **Application Domain Indexes:-**
- Application domain indexes are custom indexes that are specific to the needs of your application.
- They can be based on the semantics of your data and queries.
- **Example:**
- You could create an application domain index like this:

```
CREATE INDEX idx_library_availability ON books (branch_id,
publication_date);
```

Q3-D

1. Write a note on sequence.

- A sequence is a set of integers that are generated in order on demand.
- Sequences are frequently used in databases because many applications require each row in a table to contain a unique value and sequences provide an easy way to generate them.
- Auto increment created in sequence so no need to remember field and insert data in that field with unique number.
- Sequences can be created, changed, and dropped with the following three SQL commands:
- CREATE SEQUENCE
- ALTER SEQUENCE
- DROP SEQUENCE

- Syntax :-
- Create sequence <name>
 [Start with <no>
 Increment by <no>
 Minvalue <no> / Nominvalue
 Maxvalue <no> / Nomaxvalue
 Cycle / Nocycle
 Cache / Nocache]
- A sequence definition may consist of a start value, increment value, minimum value, and maximum value.
- You can specify whether the sequence generator should stop when reaching a boundary value, or CYCLE the sequence numbers within the minimum/maximum range.
- All sequence attributes are optional and they all have default values.
- Each sequence has two columns: NEXTVAL and CURRVAL.
- NEXTVAL is use for nextvalue in sequence while CURRVAL use for current value of data.

2. What are locks? .Explain its various types.

- In Oracle PL/SQL, a LOCK is a mechanism that prevents destructive interaction between two simultaneous transactions or sessions trying to access the same database object.
- **Modes of Locking:-**
- Exclusive lock
- The mode prevents the associates resource from being shared .
- Share lock
- The mode allows the associated resource to be shared, depending on the operations involved .
- **Locking Issues:-**
- Lost Updates
- Pessimistic Locking
- Optimistic Locking
- Blocking

- Deadlocks
- Lock Escalation
- **Lost Updates :-**
- A lost update problem occurs due to the update of the same record by two different transactions at the same time.
- In simple words, when two transactions are updating the same record at the same time in a DBMS then a lost update problem occurs.
- **Optimistic and Pessimistic lock :-**
- Optimistic locking is when you check if the record was updated by someone else before you commit the transaction.
- Pessimistic locking is when you take an exclusive lock so that no one else can start modifying the record.
- **Blocking issue :-**
- Blocking issues in Oracle occur when a session holds an exclusive lock on an object and doesn't release it before another session wants to update the same data.
- This blocks the second session until the first session has done its work.
- Blocking locks can happen when a session issues an insert, update, or delete command that changes a row.
- When the change occurs, the row is locked until the session either commits the change or rolls the change back.
- **Deadlocks :-**
- A deadlock occurs when two or more sessions are waiting for data locked by each other, resulting in all the sessions being blocked.
- Oracle automatically detects and resolves deadlocks by rolling back the statement associated with the transaction that detects the deadlock.
- **Lock escalation :-**
- Lock escalation is a process that databases use to reduce the number of locks they hold.
- It occurs when a database raises the locks to a higher level on table.
- For example, if a single user locks many rows in a table, some databases automatically escalate the user's row locks to a single table.

- **Types of Locks :-**

Lock	Description
• DML locks (data locks)	• DML locks protect data • For example, table locks lock entire tables, rowlocks lock selected rows.
• DDL locks (dictionary locks)	• DDL locks protect the structure of schema objects
• Internal locks and latches	• Internal locks and latches protect internal database structures such as datafiles

Q4-A

- 1. cursor is a database object which is used to manipulate data in a row-to-row manner.
- 2. If user defined error condition exists, the call to the user defined exception is made using a raise statement.
- 3. Which is a database object that allows us to execute a batch of SQL code when a table event occurs?

Ans. Trigger

- 4. Types data type is also known as user defined data type.

Q4-B

1. Differentiate: %type v/s %rowtype.

%type	%rowtype
• Used to declare a field with the same type as • That of a specified table's column.	• Used to declare a record with the same types As found in the specified table, view or cursor.
• %TYPE is used to declare a variable or parameter whose data type is the same as a specific database column or variable.	• %ROWTYPE is used to declare a composite variable or parameter that can store an entire row of data from a table or cursor.
• You specify the source of the data type by referencing a column, variable, or another variable using the %TYPE attribute.	• You specify the source of the structure by referencing a table, cursor, or record using the %ROWTYPE attribute.

2. Define the term:- varrays, nested table.

- **Varrays :-** The varray (variable size array) is one of the three types of collections in PL/SQL (associative array, nested table, varray).
- A varray is similar to a nested table except you must specify an upper bound in the declaration
- **Nested Table :-** Nested tables are similar to PL/SQL tables, but they can be stored in the database as a column in a table or as a database object.
- Nested tables can be considered as unbounded arrays.

Q4-C

1. What is cursor? Explain its various types.

- When you execute a SQL statement from PL/SQL, the ORACLE assigns a private work area for that SQL statement is known as CURSOR.
- The PL/SQL is mechanism by which you can name that work area and manipulate the information within it. Data stored in the cursor known as Active Data Set (ADS).
- Types of cursors in oracle

There are two types of cursor available in oracle

- 1). Implicit cursor
- 2). Explicit cursor

- **Implicit cursor :-**
- The implicit cursors are automatically generated by Oracle while an SQL statement is executed.
- These are created by default to process the statements when DML statements like INSERT, UPDATE, DELETE etc. are executed.
- **Attributes of both cursor :-**
- %ISOPEN – returns true if cursor open, false otherwise.
- %FOUND – returns true if record was fetched successfully, false otherwise.
- %NOTFOUND – returns true if record was not fetched successfully, false otherwise.

- %ROWCOUNT – returns number of records processed from the cursor.
- **Explicit cursor :-**
- A cursor can also be opened for processing data through a PL/SQL block such a user-defined cursor known as explicit cursor.
- Attributes of both cursor.
- %ISOPEN – returns true if cursor open, false otherwise.
- %FOUND – returns true if record was fetched successfully, false otherwise.
- %NOTFOUND – returns true if record was not fetched successfully, false otherwise.
- %ROWCOUNT – returns number of records processed from the cursor.

2. What is exception? Differentiate: predefined v/s user defined.

- PL/SQL supports programmers to catch such conditions using EXCEPTION block in the program and an appropriate action is taken against the error condition.

There are two types of exceptions:-

Predefined	User defined
• Predefined data types, also known as built-in or system-defined data types, are provided by the database management system (DBMS) itself.	• User-defined data types are created by users or developers to meet specific requirements that cannot be fulfilled by predefined data types alone.
• These data types are part of the SQL standard or are specific to the particular DBMS you are using (e.g., SQL Server, Oracle, MySQL).	• These data types are defined and customized by users within the SQL database using CREATE TYPE or similar SQL statements.
• Common examples of predefined data types include INTEGER, VARCHAR, DATE, and BOOLEAN, among others.	• User-defined data types can be structured types (e.g., user-defined composite types, arrays, or object types) or non-structured types (e.g., user-defined domains or synonyms).

These data types are readily available for use in your database schema without any additional definition or customization.	They are typically used to abstract and encapsulate complex data structures or to enforce data integrity constraints.
Example :- No_data_found	Example:- when not_eligible then

Q4-D

1. What is procedure ? Explain with example.

- PL/SQL block is not stored in database, its compiled every time when its executed.
- To remove these restrictions, Oracle provides facility to declare 'Name Block' like Procedures, Functions, Packages and Triggers.
- The named block can be stored in the database and can be called and executed by client applications at any time.
- Procedures and functions are subprograms which store your PL/SQL block into database server of Oracle and executes when it needed.
- Procedures take zero or more parameters and return no values.
- Procedures are called as stand alone executable statements.
- Parameter in procedure :-**
 - You may pick one of the following modes for each parameter:
 - IN is the default mode for a parameter.
 - IN parameters already have a value when the procedure is run.
 - The value of IN parameters may not be changed in the body.
 - OUT is specified for parameters whose values are only set in the body.
 - IN OUT parameters may already have a value when the procedure is called, but their value may also be changed in the body.
- Syntax for procedure:**

```
CREATE OR REPLACE PROCEDURE <procedure name>
(<parameter list>) [IS/AS]
<Declaration Section>
BEGIN
<PL/SQL code>
EXCEPTION
```

<PL/SQL code>

END;

- Example :-
- create or replace procedure revstr (st in varchar2)as
 lst number;
 st1 varchar2(20);
 begin
 lst:=length(st);
 for i in reverse 1.. lst
 loop
 st1 := st1||substr(st,i,1);
 end loop;
 dbms_output.Put_line('Reverse of string is '|| st1);
 end;
- OUTPUT :- SQL> exec revstr ('DAKU');
 Reverse Of String Is UKAD

2. Consider a student table having fields like seat.no, name, and mark. Develop a trigger on the table for insert or update any records and name should be display into the uppercase.

- -- Create the student table
- CREATE TABLE student (
 seat_no varchar2(25) PRIMARY KEY,
 name varchar2(30),
 mark int(3)
);
- -- Create the trigger
- CREATE TRIGGER uppercase_name
 BEFORE INSERT ON student
 FOR EACH ROW
 BEGIN

Insert into student

Values

(:new.seat_no,upper(:new.name),:new.mark);

END;

- -- Trigger for updates
- CREATE TRIGGER uppercase_name_update
BEFORE UPDATE ON student
FOR EACH ROW
BEGIN
 UPDATE student
 SET name = upper(:new.name)
 WHERE seat_no = :new.seat_no;
END;

Q5-A

- 1. DataFile File holds the actual data.
- 2. A table IS used to logically group data together.
- 3. LGWR stands for Log Writer Process .
- 4. SQL Server IS a principal scheduler object.

Q5-B

1. Differentiate: Data files v/s Redo log files.

Data files	Redo log files
• Data files store the actual data that is managed by the database. This includes tables, indexes, and other database objects.	• Redo log files serve as a critical component for ensuring the durability and recoverability of the database.
• When you perform INSERT, UPDATE, or DELETE operations on the database, the changes are made in the data files.	• Every time a transaction is committed, the changes made by that transaction are written to the redo log files.
• Data files are typically persisted on disk and are essential for the long-term storage and retrieval of data.	• Redo log files are cyclically overwritten, but their content is persistent until they are no longer needed for recovery purposes.

In an e-commerce database, data files would store information such as customer details, product information, and order history.	In case of a sudden power failure or system crash, redo log files can be used to replay the changes and bring the database back to its previous state.
---	--

2. Define the term: oracle blocks, import.

- oracle blocks** : Oracle blocks are used to store various types of data within the database, including table data, index data, and other structural information.
- Each Oracle block has a specific size, which can vary depending on the configuration of the database, but it is typically 8KB (kilobytes).
- Import** : the term "import" typically refers to the process of bringing data from an external source or file into a database.
- This is done to populate database tables with data from an external file or to update existing data within the database.

Q5-C

1. Write a note on initialization parameter.

- The initialization parameter file is a text file that contains a list of parameters and a value for each parameter.
- The file should be written in the client's default character set. Specify values in the parameter file which reflect your installation.
- The following are sample entries in a parameter file:

```
PROCESSES = 100
OPEN_LINKS = 12
GLOBAL_NAMES = TRUE
```
- The name of the parameter file varies depending on the operating system. For example, it can be in mixed case or lowercase, or it can have a logical name or a variation on the name INIT.ORA.
- As the database administrator, you can choose a different filename for your parameter file. There is also an INITDW.ORA file which contains suggested parameter settings for data warehouses and data marts.
- Sample parameter files are provided on the Oracle Server distribution medium for each operating system.

- A distributed sample file is sufficient for initial use, but you will want to make changes in the file to tune the database system for best performance. Any changes will take effect the next time you completely shut down the instance and then restart it.
- **Database administrators can use initialization parameters to do the following:**
 - optimize performance by adjusting memory structures, for example, the number of database buffers in memory
 - set some database-wide defaults, for example, how much space is initially allocated for a context area when it is created
 - set database limits, for example, the maximum number of database users
 - specify names of files
 - Many initialization parameters can be fine-tuned to improve database performance. Other parameters should never be altered or only be altered under the supervision of Oracle Corporation Worldwide Support staff.

2. What is memory structure ? Explain in detail.

- Memory Structures to the various data structures and components within a database management system (DBMS) that are used to manage and store data in memory. Below, I'll explain some of the key memory structures used in SQL databases in detail:
- Database buffer cache
- Redo log buffer
- Shared pool
- Dictionary cache
- **Database buffer cache:-**
 - The database buffer cache is the portion of the SGA that holds copies of data blocks read from datafiles.
 - The buffers in the cache are organized in two lists: the write list and the least recently used (LRU) list.
 - The write list holds dirty buffers, which contain data that has been modified but has not yet been written to disk.
 - The LRU list holds free buffers and dirty buffers that have not yet been moved to the write list.

- Free buffers do not contain any useful data and are available for use
- **Redo log buffer :-**
- The redo log buffer is a circular buffer in the SGA that holds information about changes made to the database. This information is stored in redo entries.
- Redo entries contain the information necessary to reconstruct changes made to the database by INSERT , UPDATE , DELETE , CREATE , ALTER, or DROP operations.
- Redo entries are used for database recovery, if necessary.
- **Shared pool :-**
- Shared pool has two cache
- **Library cache**
- Executable code which connected to instance kept in library cache.
Ex. Sql or pl/sql statements
- **Dictionary cache :-**
- Dictionary cache is stored library of all oracle statements like select, insert, update etc so when user enter any query dictionary cache checks its syntax in library

Q5-D

1. Write a note on scheduler.

- Oracle Scheduler Concepts refer to the fundamental ideas and components that form the basis of Oracle Scheduler, a powerful job scheduling and automation tool within the Oracle Database. Oracle Scheduler is designed to help automate and manage a wide range of tasks, jobs, and processes within the database environment.
- **Job:**
- A job is a specific task or unit of work that you want to automate using Oracle Scheduler. Jobs can include activities like running SQL scripts, PL/SQL procedures, external scripts, or data transformations.
- **Program:**
- A program defines what a job does. It specifies the executable and arguments required to perform a particular task. Programs can include stored procedures, executable files, or scripts.

- **Schedule:**
 - A schedule defines when and how often a job should run. It specifies the frequency, start time, end time, and recurrence pattern for executing a job.
- **Job Class:**
 - Job classes are used to categorize and manage jobs. You can assign jobs to specific job classes based on their characteristics or priorities.
- **Window:**
 - A window is a specific time interval during which jobs are allowed to run. Windows are often used to control when maintenance tasks or resource-intensive jobs can be executed, ensuring they don't interfere with critical workloads.
- **Dependency:**
 - Dependency defines the relationship between jobs. You can specify that one job should only run after another job has completed successfully. Dependencies enable you to create job chains and manage complex workflows.
 - "Scheduling Jobs with Oracle Scheduler" involves using Oracle Scheduler, a feature within the Oracle Database, to automate and manage the execution of various tasks or jobs at specified times or under specific conditions.
- **Creating Jobs :**
 - To schedule jobs with Oracle Scheduler, you first need to define the jobs you want to automate. Jobs can include tasks like running SQL scripts, PL/SQL procedures, or external scripts.
- **Defining Programs :**
 - For each job, you need to specify a program that defines what the job does. The program specifies the executable or procedure to run, as well as any required arguments or parameters.
- **Setting Schedules :**
 - Once you have defined jobs and programs, you need to create schedules that determine when and how often these jobs should run. Schedules can be configured to execute jobs at specific times of the day, week, or month, and they can include options like specifying start and end times or recurring patterns

- **Monitoring and Notifications:**
- Oracle Scheduler provides tools for monitoring the execution of scheduled jobs. You can view job status, logs, and performance metrics.
- **Job History:**
- Oracle Scheduler maintains a history of job executions, allowing administrators to track the progress of jobs, view job status, and diagnose problems.
- **Security and Access Control:**
- Access control mechanisms ensure that only authorized users can create, modify, or execute scheduled jobs.
- **Administering Oracle Schedule**
- **Job Management:**
- Administrators are responsible for creating, modifying, and deleting jobs. This includes defining the job type (e.g., SQL script, PL/SQL procedure), setting job attributes, specifying the program to execute, and assigning the job to schedules.
- **Programs and Executables:**
- Administrators manage programs and executables associated with jobs. This involves defining the programs, specifying executable files or PL/SQL procedures, and configuring program attributes.
- **Schedules:**
- Administering Oracle Scheduler involves the creation, modification, and deletion of schedules.
- **Dependency Management :**
- Administrators establish dependencies between jobs when needed. Dependencies define the order in which jobs are executed and ensure that specific conditions are met before a job starts.
- **Window Configuration :**
- Windows, which specify time intervals during which jobs are allowed to run, are set up and managed by administrators.
- **Resource Allocation :**
- Resource plans and job classes are configured to manage resource allocation to jobs.

2. What is instance architecture ? Explain.

- Instance Architecture the overall structure and components that make up a running instance of a DBMS.
- The term "instance" typically represents a single, independent copy of a database management system, which is capable of serving multiple users and processing database operations.
- When a database is started on a database server, Oracle allocates a memory area called the System Global Area (SGA) and starts one or more oracle processes.
- This combination of the SGA and the oracle processes is called an Oracle Instance
- **The instance and the Database :-**
- This is called Mounting the database. The database then ready to be Opened, which makes it accessible to authorized users. Only the database administrator can start up an instance and open the database.
- If a database is open, then the database administrator can shut down the database so that its closed
- Security for database startup and shutdown is controlled through connection to oracle with administrator privileges
- Normal users do not have control over the current status of an oracle database.
- **Instance and database startup :-**
- Start an instance
- Mount the database
- Open the database
- **Database and instance shutdown :-**
- Open the database
- Close the database
- Unmount the database
- Shut down the instance

- **An Oracle instance is build up by considering following aspects :-**
 1. Database Process
 2. Memory Structure
 3. Data Files
- **Database Process :-**
- The Oracle RDBMS uses two types of processes , user process and oracle process (called background process)
- **User processes :**
- Its user's connection to the RDBMS system. It communicates user input with the Oracle server process-by using oracle program interface.
- Its also used to display the information requested by the user.
- **Oracle processes :-**
- It performs the function for users which divided into two processes :
 - A. server processes
 - B. Background processes
- **Server process:-**
- Oracle creates server process to handle the request of user processes. This server creates for the for the following
 - Run sql statements issued through the application
 - Read necessary data blocks from data files on disk
 - Return the result in such a way that the application can process the information
- **Background process :-**
- To maximize performance and accommodate many users, a multiprocess Oracle system uses some additional oracle processes called background processes.
- Background process in Oracle are follow :

➤ Database Writer Process (DBW) ➤ Log Writer Process (LGWR) ➤ Checkpoint Process (CKPT) ➤ System Monitor Process (SMON)	➤ Process Monitor Process (PMON) ➤ Archiver Processes (ARCn) ➤ Recoverer Process (RECO) ➤ Dispatcher ➤ Server
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JANUARY – 2021

CS-15: RDMS USING ORACLE

Q-1(A). Answer the following. (04)

1. Give the full form of DCL - **Data Control Language**.
2. Describe title and repheader SQL *Plus formatting command.

A. The title command defines the top title.

Example: title center 'system user'

- ☒ The **repheader** command defines the report header.
- 3. Give the full form of DQL - **Data Query Language**.
- 4. Give the full form of SQL - **Structure Query Language**.

Q-1(B). Answer in Brief. (02)

1. What is DBMS?

A. DBMS stand for Database Management System

- ☒ DBMS used to manage your data like insert, update and delete etc.

Q-1(C) Answer in Detail. (03)

1. Write a note on Normalization.

- Normalization is a database design technique which organizes tables in a manner that reduces redundancy and dependency of data.

- ☒ It divides larger tables to smaller tables and links them using relationships.

- ☒ There are three types of rules.

☒ **1NF (First Normal Form) Rules**

- Each table cell should contain a single value.

- Each record needs to be unique.

☒ **2NF (First Normal Form) Rules**

- Rule 1- Be in 1NF
- Rule 2- Single Column Primary Key

☒ **3NF (First Normal Form) Rules**

- Rule 1- Be in 2NF
- Rule 2- Has no transitive functional dependencies.
- To move our 2NF table into 3NF, we again need to again divide our table.
- There are no transitive functional dependencies, and hence our table is in 3NF.
- In Table 3 Salutation ID is primary key, and in Table 1 Salutation ID is foreign to primary key in Table 3.

Q-1(D) Answer the following.

(05)

1. Difference between DBMS v/s RDBMS.

DBMS	RDBMS
Relationship between two tables or files are maintain programmatically.	Relationship between two tables or file can be specified at the time of table creation
Does not support client/server architecture.	Supports client/server architecture.
Does not support distributed database.	Supported distributed database.

No security of data.	Multiple security levels; 1. Login at OS level 2. Command level security 3. Object level security
Each table is given extension.	Many tables are grouped in one database in RDBMS.
Example: foxpro	Example: oracle

Q-2(A). Answer the following.

(04)

1. Give the full form of DML – **Data Manipulation Language**
 2. Give the full form of DDL – **Data Dictionary Language**
 3. Give the full form of RDBSM – Relational Database Management System
 4. Describe set linesize and set heading sql *plus formatting command.
- A. To avoid this problem, you can change the oracle default settings to increase width of the lines, using the set commands.
Example: set linesize 100
- ☒ The new heading will remain in effect until you enter different headings, reset each column's format, or exit from SQL *Plus.
Example: set heading on/off

Q-2(B). Answer the following.

(02)

1. **What is BCNF?**
- A. Boyce Code Normal Form
- ☒ Boyce Code Normal Form is based on functional dependencies that take into account all candidate keys in a relation.

- ⊗ However, BCNF also has additional constraints compared with the general definition of 3NF.

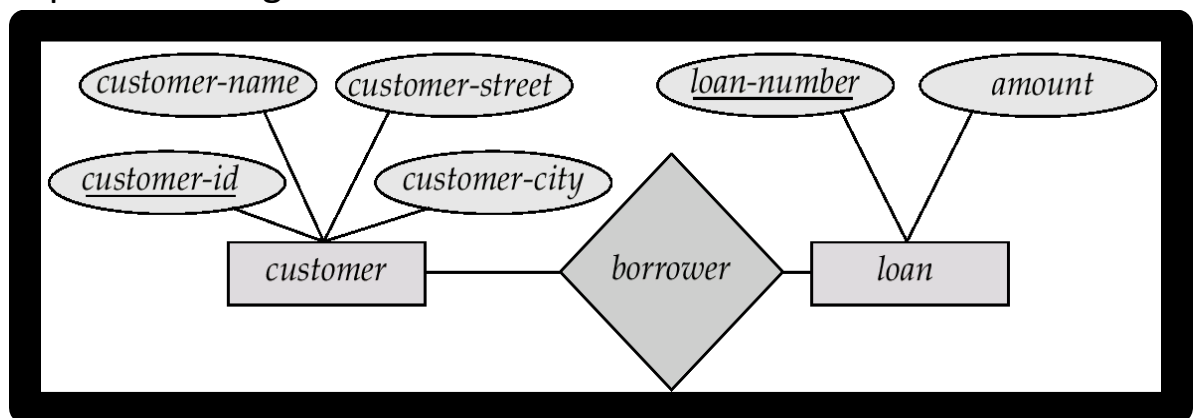
Rules for BCNF:

- ⊗ Rule 1: The table should be in 3rd normal form
- ⊗ Rule 2: X should be a superkey for every functional dependency (FD) $X \rightarrow Y$ in a given relation.

Q-2(C) Answer in Detail.

(03)

1. Explain E R Diagram.



- **Rectangles** represent table sets.
- **Diamonds** represent relationships between tables.
- **Lines** link attributes represent connection between tables with relationship and field.
- **Ellipses** represent table fields
- **Underline** indicates primary key attributes

Q-2(D) Answer the following.**(05)**

1. Difference between SQL v/s SQL *Plus.

SQL	SQL *Plus
SQL is a language for communicating with the oracle server to access data	SQL *PLUS recognizes SQL statements and sends them to Server
SQL is based on American National Standards Institute(ANSI)	SQL *PLUS is the Oracle proprietary interface for execution SQL statements
SQL manipulates data and table definition in the database	SQL *PLUS does not allow manipulation of values in the database
SQL is entered into the SQL buffer on one or more lines	SQL *PLUS is entered one line at a time, not stored in the SQL buffer
SQL does not have continuation character	SQL *PLUS uses a dash(-) as a continuation character if the command is longer than one line
SQL uses a termination character to execute commands immediately	SQL *PLUS does not require termination characters ; executes commands immediately

SQL uses functions to perform some formatting	SQL *PLUS uses commands to format data
---	--

Q.3 (A) Answer the following:-

1. A _____ is a database object that holds users data.

Ans. Table

2. All table columns belong to a single record.[True/False].

Ans. False

3. The update command is used to change or modifying data values in a table.[True/False]

Ans. True

4. A _____ keys represents relationship between tables.

Ans. Foreign keys

(B) Answer in brief

What is candidate key?

Ans.

A **candidate key** is a key that uniquely identifies rows in a table. Any of the identified candidate keys can be used as the table's primary key. Candidate keys that are not part of the primary key are called alternate keys.

One can describe a candidate key as a super key that contains only the minimum number of columns necessary to determine uniqueness.

(c) Answer in detail

Describe abs(), least(), months_between() functions.

□ abs(): returns absolute value

e.g:- Select abs(-12) from dual;

□ Least : returns smallest value out of list of nos.

e.g:- Select least(10,20,30) from dual;

□ MONTHS_BETWEEN(x,y) returns the number of months between x and y. If x is before y on the calendar, the number returned is positive; otherwise the number is negative.

e.g:- **SELECT** MONTHS_BETWEEN('25-MAY-2009',
'15-JAN-2012') **FROM** dual;

result:-

-31.677419

(D) Write a note on select statement.

The SELECT statement is a SQL statement that is used to retrieve data from a database. It is one of the most commonly used SQL statement

The syntax for a SELECT statement is as follows:

```
SELECT column_list FROM table_name  
[WHERE condition]
```

[ORDER BY column_name]

Column list

The column_list specifies the columns that you want to retrieve from the table. The * in the column_list indicates that all of the columns from the table should be retrieved.

Table name

The table_name specifies the name of the table that you want to retrieve data from.

WHERE condition

The WHERE clause is used to filter the rows that are returned. The condition can be a simple comparison, such as country = 'USA', or it can be a more complex expression.

ORDER BY clause

The ORDER BY clause is used to sort the rows that are returned. The column name that you specify in the ORDER BY clause determines the sort order. The default sort order is ascending, but you can also specify descending order by adding the DESC keyword after the column name.

Q4 (A) Answer the following:-

1. A _____ constraint must be specified as a logical expression that evaluate true or false.

Ans. Check

2. A _____ is a small Oracle Worktable, which consists of only one row and one column.

Ans. Dual

3. The _____ clause imposes a condition on the group by clause.

Ans. Having

4. A subquery is also termed as nested query. [true/false]

Ans. True

(B) Answer in brief:-

Describe initcap(), soundex() functions

□ Initcap() : converts first character of all words into capital and remaining into small

e.g:- Select initcap('welcome to hjd college') from dual;

☒ soundex():-

The SOUNDEX() function returns a string that contains the phonetic representation of a string.

Syntax:-

SOUNDEX(expression)

the The SOUNDEX() function is useful for comparing words that sound alike but spelled differently in English.

expression is a literal string or an expression that evaluates to a string.

The SOUNDEX() function will return a string, which consists of four characters, that represents the phonetic representation of the expression.

e.g:- SELECT
 SOUNDEX('see') see,
 SOUNDEX('sea') sea
FROM
 dual;

Result:-

SEE SEA

---- ----

S000 S000

Because both words sound the same, they should receive the same Soundex value.

(C) Explain GRANT, REVOKE , ROLLBACK command

□ **GRANT** : Grant is use to grant privileges on tables, view, procedure to other users or roles

e.g:-

-> grant select, update, insert on emp to scott;

-> grant all on emp to scott;

□ **REVOKE** :Use to take back privileges granted to other users and roles.

e.g:-

revoke select, update, insert on emp from scott;

□ **ROLLBACK** : the **ROLLBACK statement** is used to undo the work performed by the current transaction or a transaction that is in doubt.

Syntax:

Rollback [work] [to [savepoint] savepoint_name | force 'string']

Parameters or Arguments

WORK

Optional. It was added by Oracle to be SQL-compliant. Issuing the ROLLBACK with or without the WORK parameter will result in the same outcome.

TO SAVEPOINT savepoint_name

Optional. The ROLLBACK statement undoes all changes for the current session up to the savepoint specified by *savepoint_name*. If this clause is omitted, then all changes are undone.

FORCE 'string'

Optional. It is used to force the rollback of a transaction that may be corrupt or in doubt. With this clause, you specify the transaction ID in single quotes as *string*. You can find the transaction ID in the system view called DBA_2PC_PENDING.

e.g:-

□ Rollback to savepoint s1;

(D) Write a note on join.

A JOIN clause is used to combine rows from two or more tables, based on a related column between them.

Different Types of SQL JOINS:-

1.Inner Join

2. Outer Join

- a. Left outer Join
- b. Right outer Join
- c. Full outer Join

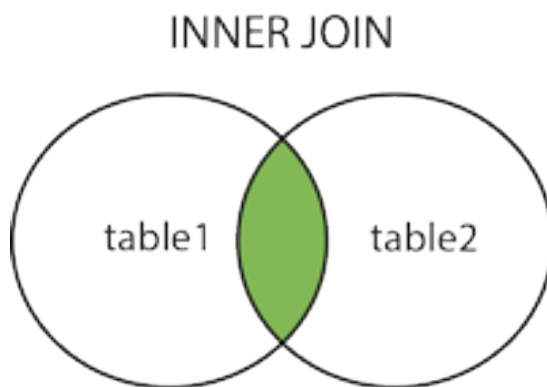
3. Self Join

4. Cross Join

1. SQL INNER JOIN Keyword

The INNER JOIN keyword selects records that have matching values in both tables.

This join is also known as equi join

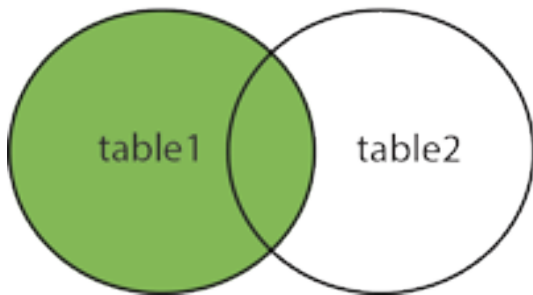


2. Outer Join

a. SQL LEFT JOIN Keyword

The LEFT JOIN keyword returns all records from the left table (table1), and the matched records from the right table (table2).

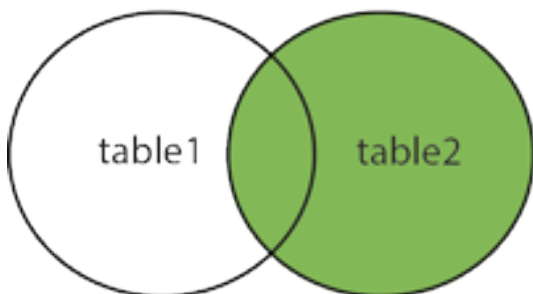
LEFT JOIN



b. SQL RIGHT JOIN Keyword

The RIGHT JOIN keyword returns all records from the right table (table2), and the matched records from the left table (table1).

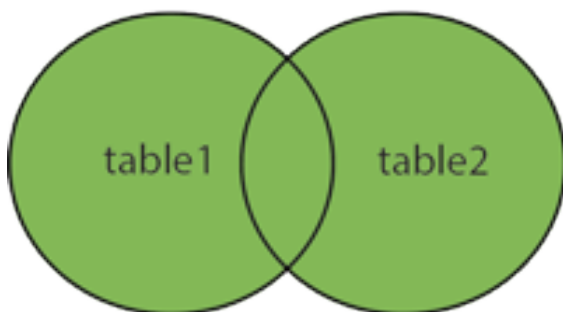
RIGHT JOIN



c. SQL FULL OUTER JOIN Keyword

The FULL OUTER JOIN keyword return all records when there is a match in either left (table1) or right (table2) table records.

FULL OUTER JOIN



3. SQL Self JOIN

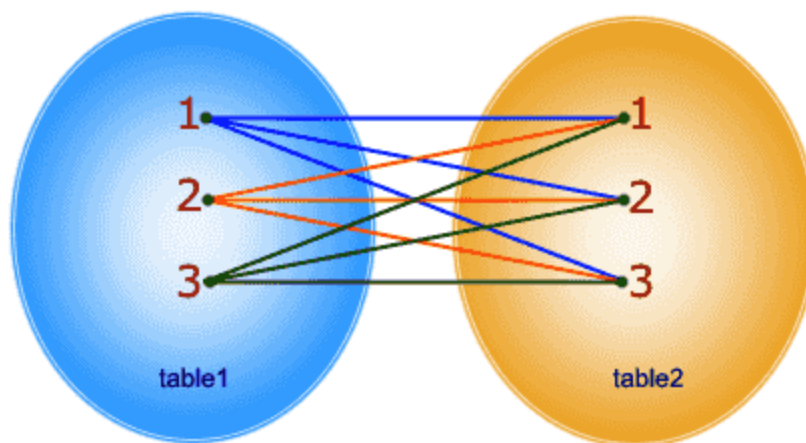
A self JOIN is a regular join, but the table is joined with itself.

4. Cross join

In this join there is no condition given so records multiply by each other table

Like if one table have 3 records and other table have 4 and we apply cross join then result will be 12 rows display

```
SELECT * FROM table1 CROSS JOIN table2;
```



In CROSS JOIN, each row from 1st table joins with all the rows of another table.
If 1st table contain x rows and y rows in 2nd one the result set will be $x * y$ rows.

Q-5 (A) Answer the following

(04)

- 1) A view is used to only look and table data and nothing else. The view is called a **Read Only View**.
- 2) In a **B-Tree** index, you walk the branches until you get to the node that has the data you want to use.
- 3) A **Database Link** object is telling oracle how to get from one database to another.
- 4) Two database operations wait for each other to release a lock is known as **Deadlock**.

(B) Answer in brief.

(02)

1) What are latches?

- Ans.**
- > A latch is a low-level internal lock used by oracle to protect memory structure.
 - > Latches are serializations mechanisms that protect areas of oracle's memory.
 - > The latches within oracle can be broken down into short and long wait latches.
 - > The purpose of latches is to manage simultaneous access to shared data structures so that only one process can access the structure at a time.

(C) Answer in detail.

(03)

1) Explain cluster.

- Ans.**
- > Clusters in SQL are used to store data that is from different tables in the same physical data blocks.
 - > They are used if records from those tables are frequently queried together.
 - > Each cluster stores tables data and maintains a clustered index to sort data.
 - > Columns within the cluster index are called clustered keys.
 - > These determine the physical placement of rows within the cluster.
 - > Cluster key is usually a foreign key of one table that references the primary key of another table in cluster.
 - > The first cluster is created. Then cluster index is created on cluster key columns.
 - > After cluster key index is created, data can be entered into tables stored in clusters.

=> Advantages of clusters :

- > Disk I/O is reduced.
- > Access time improves for joins of clustered tables.

=> Disadvantages of clusters:

- > Columns that are often updated are not good candidates for cluster key.
- > Reduces performance of INSERT statements as compared to storing the table separately with it's own index.

EXAMPLE:

```
create table branch_master  
(branch_code varchar(10) PRIMARY KEY,  
branch_name varchar(15) );
```

-> Now, Consider the above created table and let's create the cluster for the same.

Creating a Cluster :

```
create cluster branch_info  
(branch_code varchar(10) );
```

(D) Answer the following. (05)

1) Write a note on sequence.

Ans. -> A sequence is a set of integers that are generated in order on demand.

-> Sequences are frequently used in databases because many applications require each row in a table to contain a unique value and sequences provide an easy way to generate them.

-> Sequences can be created, changed, and dropped with the following three SQL commands:

- 1) CREATE SEQUENCE
- 2) ALTER SEQUENCE
- 3) DROP SEQUENCE

Syntax:-

Create sequence <name>

[Start with no

Increment by

Minvalue / Nominvalue

Maxvalue / Nomaxvalue

Cycle / Nocycle

Cache / Nocache]

->A sequence definition may consist of a start value, increment value, minimum value, and maximum value.

-> All sequence attributes are optional and they all have default values.

-> Each sequence has two columns: NEXTVAL and CURRVAL.

-> NEXTVAL is use for nextvalue in sequence while CURRVAL use for current value of data.

EXAMPLE:

-----create sequence-----

Create sequence stud_seq

Start with 10

Increment by 2

Minvalue 5

Maxvalue 30

Cycle

Cache 2;

-----create table-----

Create table stud

(sno number(3),
Sname char(30),
Seqno number(3),
Gender char(10));

-----insert records-----

Insert into stud values(&sno,'&sname',stud_seq.nextval,'&Gender');

Q-6 (A) Answer the following. (04)

- 1) Oracle used a method called **Multi-version concurrency control (MVCC)** to implement concurrency control.
- 2) The oracle engine automatically locks table data while executing SQL statement. [true/false]

Ans. -> True

- 3) The select.. for update can be used with set operators and group by clause. [true/false]

Ans. -> False

- 4) Latches request can be made two modes **“willing to wait”** or **“no wait”**.

(B) Answer in brief. (02)

- 1) what is synonym?

Ans. -> A synonym is an alternative name for objects such as tables, views, sequences, stored procedures, and other database objects.

-> Synonyms is dependent object.

-> A synonym is an alias for a schema object.

SYNTAX:

Create Synonym `Synonym_Name` for `Table_name`;

(C) Answer in detail.

(03)

1) Explain Deadlock.

Ans. -> A deadlock occurs when two or more sessions are waiting for data locked by each other, resulting in all the sessions being blocked.

-> Oracle automatically detects and resolves deadlocks by rolling back the statement associated with the transaction that detects the deadlock.

Resolving Deadlock:

1) Ask the session getting deadlock error ORA-00060 to issue either **commit** or **rollback**.

2) Ask the waiting session to kill the SQL/transaction.

-> The following script can be used to identify deadlocks in the database.

-> The query depends upon objects that are created by the script

`$ORACLE_HOME/rdbms/admin/dbmslock.sql`

Avoid deadlock:

->if you want to avoid the deadlock, try to use the clause skip locked.

(D) Attempt the following.

(05)

1) Write a note on View.

Ans. ->A view is a virtual table that stores only definition in the form of SQL statement.

-> It is used to hide information from other user.

Feature of view:

-> Creating view does not take any storage space in oracle.

-> The maximum number of columns that can be defined in a view is 1000.

-> When new records are inserted/updated/deleted in base table, the changes are automatically reflected in the view.

-> View once defined, can be used in Joins and Sub-query also.

Types of view:

1) Read only view

2) Updateable view

Syntax of view :-

```
CREATE [OR REPLACE] [FORCE] VIEW<View Name>  
AS  
SELECT<Fields> FROM <Table>WHERE <condition >
```

EXAMPLE:

1) Read only View:

Create or replace view vemp as

Select ename,sal from emp where sal>2000

With read only;

Insert into vemp values('SCOTT',3000);

2) Updatable View:

Create or replace view vemp as

Select ename,sal from emp where sal>1000;

Insert into vemp values('SCOTT',3000);

Q.7(a) write the following.

(1) Give full form of PL/SQL.

Ans. PL/SQL stands for Procedural Language/SQL

Or

Procedural Language extensions to SQL.

(2) Code blocks starts with a _____ section.

Ans. Declaration section

(3) A _____ object holds procedure, functions, variable, constants, cursors and exceptions.

Ans. Package

(4) An appropriate action is taken against the error condition known as _____

Ans. Exception block

(b) Answer in brief.

(1) What is %Rowtype?

Ans.

- Using %ROWTYPE attributes, a record variable can be declared implicitly based the structure of a table or query
- In implicit defined record, we don't have to describe each field in the record definition
- It mean oracle select data type automatically for declaring %ROWTYPE attribute
- Example :-
 - salary emp%rowtype

(c) Answer in detail.

(1) Explain a PL/SQL block structure.

Ans. PL/SQL block structure :

- Any language is useless if it has no programming power.
- By writing simple SQL statement, we cannot fulfill all our requirements.

- As result PL/SQL is required for the purpose of programming.
- PL/SQL stands for Procedural Language/SQL
- PL/SQL is bloc structure language. Each block divided into three sections
 - Declaration Section (Optional)
 - Execution Section
 - Exception Section (Optional)

Declaration Section :

- This is section is start with keyword DECLARE
- Variables and constants are declared, initialized within this section.
- You must have to declare variables and constants in declarative block before using them in procedural statement.
- Declare Variables and Assigning values: You can define variable name, data type of a variable and its size.
- Data type can be: CHAR, VARCHAR2, DATE, NUMBER, etc
- This section is optional if no variables or constraints to declare

Execution Section :

- This section is mandatory in PL/SQL
- This section begins with keyword BEGIN
- This section can have another PL/SQL block inside it
- Execution Section contains SQL – PL/SQL executable statements
- This section contains conditional statements (if...else), looping statements (for, while) and Branching Statements (goto) etc.

Exception Section:

- This is an Optional section
- This section start with keyword EXCEPTION
- PL/SQL easily detects user defined or predefined error condition.
- PL/SQL is famous for handling errors in smart way by giving suitable user friendly messages.
- Errors can be rise due to wrong syntax, bad logical or not passing a validation rules.

syntax :

DECLARE

<variable / constant declaration>

BEGIN

<statements>

```
EXCEPTION
    <statements>
END;
```

Example:

```
declare
    nam emp.ename%type;
begin
    nam:='&nam';
    select ename into nam from emp where ename=nam;
    dbms_output.put_line('employee found' || nam);
exception
    when no_data_found then dbms_output.put_line('emp not
found');
end;
```

(d) Answer the following

(1) write a note on Package.

Ans. Package :

- A package is an object, which holds other objects like procedure, functions, cursor, exception etc. within it
- Components of Package :-

Package specification

Package body

○ Package specification :-

- Contains Name of the package
- Contains declaration of Procedure, Function, Cursors, Exception etc.
- The objects declared in a package header are visible outside the package
- It does not contain any code for procedures / functions
- Object must be declared in this section before it is referenced

○ Syntax :

CREATE [OR REPLACE] PACKAGE <package name>

{AS / IS }

Public_Variable_declarations

Public_Exception_declarations

Public_Cursor_declarations

Function_specifications

Procedure_specifications

END [package name];

Package Body :

- Package body contains the definition of public objects that are declared in the specification
- Package body can also have other objects, which are private to the package
- If package header does not contain any procedure / functions then, package body is optional
- Private objects are defined in package body only and they are not visible / accessible outside the package.
- The variables declared in the package specification can be accessed by any procedure / function within the Package
- Package object referred as

EXECUTE <package Name>.<Object Name>

or

CALL <package Name>.<Object Name>

PACKAGE BODY SYNTAX :-

CREATE [OR REPLACE] PACKAGE BODY <package name>

{AS / IS }

Private_Variable_declarations

```
Private_Exception_declarations
Private_Cursor_declarations
Function_specifications
Procedure_specifications
END [package name];
```

Example:

-----package specification-----

create or replace package pack as

procedure pro1(n1 number,n2 number,n3 number);

procedure leap (y number);

end pack;

-----package body-----

create or replace package body pack as

procedure pro1(n1 number,n2 number,n3 number) as

vsum number(10);

begin

vsum:=n1+n2+n3;

dbms_output.put_line('sum: ' || vsum);

**dbms_output.put_line('greatest value:
' || greatest(n1,n2,n3));**

end pro1;

```
procedure leap(y number)as
begin
    if mod(y,4)=0 then
        dbms_output.put_line(y || ' is a leap year');
    else
        dbms_output.put_line(y || ' is not a leap year');
    end if;
end leap;
end pack;
```

```
-----execute package-----
execute pack.pro1(10,20,30);
execute pack.leap(2024);
```

Q.8 (a) Answer the following

(1) The _____ attribute to declare variables bases on definitions of columns in a table.

Ans. %type

(2) The _____ statement causes the loop to skip the remainder of its body and immediately retest its condition prior to reiterating.

Ans. Continue

(3) The oracle engine opened a cursor for its internal processing it is known as _____.

Ans. implicit cursor

(4) A table within a table is known as _____.

Ans. nested table

(b) Answer in brief.

(1) What is explicit cursor?

Ans. A cursor can also be opened for processing data through a PL/SQL block such a user-defined cursor known as explicit cursor

Syntax:

CURSOR cursor_name IS select_statement;

(c) Answer in detail.

(1) Explain Trigger.

Ans.

TRIGGER :

- Triggers are the programs that are executed automatically in response to a change in the database.
- Oracle allows special types of procedures that are automatically executed when events like INSERT/ UPDATE/ DELETE occurs. These event procedures called “DATABASE TRIGGERS”
- You may not use COMMIT, ROLLBACK and SAVEPOINT statement within trigger blocks.
- Types of triggers
 - Before update statement level
 - After insert row level
 - Before delete row level
- Before update statement level
 - In this type, trigger fires when any kind of updates occur
 - For ex. If you write query of update and this type of trigger fired on your table the changes occur before update any data
- After insert row level
 - In this type, trigger fires when row insert in database table
 - For ex. If you insert any data in your database table and trigger is fired in your table then after inserting row trigger run
- Before delete row level

- In this type, trigger fires before delete any record in table row
- For ex. If you are delete any records in your database table this trigger fires automatically and make changes as per trigger programmed

Syntax

```

CREATE [OR REPLACE] TRIGGER <name>
[BEFORE / AFTER / INSTEAD OF]
[DELETE / INSERT / UPDATE [OF <col1>,<col2>..]]
ON <table / view name>
FOR EACH ROW [WHEN <condition>]
DECLARE
    <VAR>
BEGIN
    PL/SQL STATEMENT
EXCEPTION
    EXCEPTION
END;
```

TRIGGER PREDICATES :

- When the trigger is created for more than one DML operation, one can use conditional predicates within the trigger body to

execute specific blocks of code, depending on the types of statement that fires the trigger.

- Inserting :
TRUE if the triggering statement is an INSERT
- Deleting:
TRUE if the triggering statement is an DELETE
- Updating:
TRUE if the triggering statement is an UPDATE

:OLD AND :NEW

- The :new refers to a record containing the new, post change data.
- The :old refers to a record containing the old, pre-change data.
- The trigger can have visibility to both old and new values.
- :NEW can be modified only in a before row-level trigger, and :OLD can never be modified but only read from

- Insert:
:old :-

Undefined all fields are null

:new :-

Values that will be inserted when the statement is complete

- Update:

:old :-

Original values for the row before update

:new :-

New values that will be updated when the statement is complete

- Delete:

:old :-

Original values before the row is deleted

:new :-

Undefined all fields are null

Example:

create or replace trigger msg

before insert or delete or update on stud

for each row

begin

if inserting then

dbms_output.put_line('Record inserted in stud');

elsif deleting then

dbms_output.put_line('Record deleting in stud');

elsif updating then

dbms_output.put_line('Record updating in stud');

end if;

end;

(d) Answer the following.

(1) Difference between Procedure V/S Function.

Function	Procedure
Functions always return a value after the execution of queries.	The procedure can return a value using “IN OUT” and “OUT” arguments.
In SQL, those functions having a DML statement can not be called from SQL statements. But autonomous transaction functions can be called from SQL queries.	A procedure can not be called using SQL queries.
Each and every time functions are compiled they provide output according to the given input.	Procedures are compiled only once but they can be called many times as needed without being compiled each time.
A Function can not return multiple result sets.	A procedure is able to return multiple result sets.
The function can be called using Stored Procedure.	While procedures cannot be called from function.
A function used only to read data.	A procedure can be used to read and modify data.
The return statement of a function returns the control and function’s result value to the calling program.	While the return statement of the procedure returns control to the calling program, it can not return the result value.

The function does not support try-catch blocks.	Procedure supports try-catch blocks for error handling.
A function can be operated in the SELECT statement.	While it can't be operated in the SELECT statement.
Functions do not permit transaction management.	It allows transaction management.
In functions, we can use only a table variable. Temporary tables can not be created in function.	In procedures, we can use temporary tables or table variables to store temporary data.

Q.9 (A) Answer The Following.

(1) Give Full Form of SGA.

=> System Global Area

(2) Give Full Form of SMON.

=> System MONiter

(3) Give Full Form of DBWn.

=> DataBase Writer Process

(4) Give Full Form of RBS.

=> Rollback Segment

(B) Answer in Brief.

(1) What is Control File?

=> The control file of database is binary file that contains a great deal

of database information The control file contains name, data about the database log file.

Oracle allows you to maintain duplicate copies with the help of control files.

Its good if more then one copies stored in different disks to protect your database information.

(C) Answer in Detail.

(1) Explain Oracle Block.

Ans. => PL/SQL block structure :-

- ❑ Any language is useless if it has no programming power.
- ❑ By writing simple SQL statement, we cannot fulfill all our requirements.
- ❑ As result PL/SQL is required for the purpose of programming.
- ❑ PL/SQL stands for Procedural Language/SQL.
- ❑ PL/SQL is bloc structure language. Each block divided into three sections.
- ❑ Declaration Section (Optional).
- ❑ Execution Section.
- ❑ Exception Section (Optional).

Declaration Section :-

- ❑ This section starts with the keyword DECLARE.
- ❑ Variables and constants are declared, initialized within this section.
- ❑ You must have to declare variables and constants in a declarative block before using them in a procedural statement.
- ❑ Declare Variables and Assigning values: You can define variable name, data type of a variable and its size.
- ❑ Data type can be: CHAR, VARCHAR2, DATE, NUMBER, etc.
- ❑ This section is optional if no variables or constraints to declare .

EXECUTABLE SECTION :-

- ❑ This section is mandatory in PL/SQL.
- ❑ This section begins with the keyword BEGIN.
- ❑ This section can have another PL/SQL block inside it.
- ❑ Execution Section contains SQL – PL/SQL executable statements.
- ❑ This section contains conditional statements (if...else), looping

statements (for, while) and Branching Statements (goto) etc.

EXCEPTION SECTION :-

- This is an Optional section.
- This section start with keyword EXCEPTION.
- PL/SQL easily detects user defined or predefined error condition.
- PL/SQL is famous for handling errors in smart way by giving suitable user friendly messages.
- Errors can be rise due to wrong syntax, bad logical or not passing a validation rules.

Plsql block structure syntax :-

Syntax

■ DECLARE

■ <variable / constant declaration>

■ BEGIN

■ <statements>

■[EXCEPTION

■<statements>]

■END;

(D) Answer the Following.

(1) Write a Note on Managing Automated Database Maintenance Tasks.

=> Managing Automated Database Maintenance Task in Oracle

- This is to refer the process of using Oracle Database's built-in automation features and tools to efficiently and proactively maintain the health, performance, and reliability of an Oracle database.
- Oracle provides a range of automated tasks and features to simplify routine maintenance activities that database administrators (DBAs) typically perform manually.
- **These tasks include:**

Backup and Recovery:

- Oracle offers tools like Recovery Manager (RMAN) to automate database backups, ensuring that data can be recovered in case of

a failure.

Patching and Upgrades:

- Oracle provides automated patching and upgrade solutions, such as Patch Set Updates (PSUs) and Critical Patch Updates (CPUs), to keep the database software up-to-date with the latest security and performance enhancements.

Statistics Gathering:

- Regularly collecting and refreshing database statistics is crucial for query optimization. Oracle can automatically gather statistics on database objects.

Performance Monitoring and Tuning:

- Oracle's Automated Database Diagnostic Monitor (ADDM) analyzes database performance and provides recommendations for improving it.

Space Management:

- Automated features help manage tablespace, control file, and redo log file space.

Task Scheduling:

- Oracle allows DBAs to schedule routine maintenance tasks, making it easier to execute them during non-peak hours.

Alerts and Notifications:

- Automated systems can generate alerts and notifications when predefined conditions or thresholds are met, allowing for proactive problem resolution.

Resource Management:

- Oracle Database can automatically allocate and manage system resources to optimize performance for various workloads.

Security Updates:

- Automatic security updates help protect the database against vulnerabilities.

Q.10 (A) Answer The Following.

(1) Give Full Form Of PGA.

=> **P**rogram **G**lobal **A**rea

(2) Give Full Form Of PMON.

=> **P**rocess **MON**iter

(3) Give Full Form Of CKPT.

=> **C**heck **P**oint **P**rocess

(4) Give Full Form Of LGWR.

=> **L**o**G** **W**Riter **P**rocess

(B) Answer in Brief.

(1) What is Rollback Segment.

=> In the ORACLE database, the rollback segment is a structure that stores previous data during transactions.

- To support the rollback facility we have a rollback segment in the ORACLE database.

(C) Answer in Detail.

(1) Explain Tablespace.

=> The Oracle tablespace and how Oracle uses tablespaces to logically store the data in the database.

What is an Oracle Tablespace?

- Oracle divides a database into one or more logical storage units called tablespaces.
- Each tablespace consists of one or more files called datafiles. A datafile physically stores the data objects of the database such as tables and indexes on disk.
- In other words, Oracle logically stores data in the tablespaces and physically stores data in datafiles associated with the corresponding tablespaces.

By using tablespaces, you can perform the following operations :-

- Control the storage size allocated for the database data.

- Grant specific space quotas to the database users.
- Control the availability of data by taking tablespaces online or offline (more on this later).
- Improve the performance of the database by allocating data storage across devices.
- Perform partial database backup or recovery.

Read-Only Tablespaces :-

- The read-only tablespaces allow Oracle to avoid performing backup and recovery of large, static parts of a database. Because Oracle doesn't update the files of a read-only tablespace, you can store the files on the read-only media.
- Oracle allows you to remove objects such as tables and indexes from a read-only tablespace. However, it does not allow you to create or alter objects in a read-only tablespace.

- When you create a new tablespace, it is in the read-write mode. To change a tablespace to read-only tablespace, you use the ALTER TABLESPACE command with the READ ONLY option.

More on Oracle Tablespaces :-

- **Create tablespace** – show you how to create a new tablespace in the database.
- **Drop tablespace** – describe the steps of removing a tablespace from the database.
- **Extend tablespace** – how to extend the size of a tablespace.
- **Temporary tablespace** – manipulate temporary tablespace effectively.
- **Tablespace group** – how to use the tablespace group more effectively to optimize internal Oracle operations.

(D) Answer the Following.

(1) Write A Note On Oracle Scheduler Concept.

=> Oracle Scheduler Concepts refer to the fundamental ideas and components that form the basis of Oracle Scheduler, a powerful job

scheduling and automation tool within the Oracle

Database. Oracle Scheduler is designed to help automate and manage a wide range of tasks, jobs, and processes within the database environment.

Job :-

- A job is a specific task or unit of work that you want to automate using Oracle Scheduler. Jobs can include activities like running SQL scripts, PL/SQL procedures, external scripts, or data transformations. Jobs are at the core of Oracle Scheduler.

Program :-

- A program defines what a job does. It specifies the executable and arguments required to perform a particular task. Programs can include stored procedures, executable files, or scripts. A job references a program to execute a specific task.

Schedule :-

- A schedule defines when and how often a job should run. It specifies the frequency, start time, end time, and recurrence pattern for executing a job. Schedules allow you to control the timing of job execution.

Job Class :-

- Job classes are used to categorize and manage jobs. You can assign jobs to specific job classes based on their characteristics or priorities. Job classes help in resource allocation and prioritization of job execution.

Window :-

- A window is a specific time interval during which jobs are allowed to run. Windows are often used to control when maintenance tasks or resource-intensive jobs can be executed, ensuring they don't interfere with critical workloads.

Dependency :-

- Dependency defines the relationship between jobs. You can specify that one job should only run after another job has completed successfully. Dependencies enable you to create job chains and manage complex workflows.

Scheduling Jobs with Oracle Scheduler

- "Scheduling Jobs with Oracle Scheduler" involves using Oracle Scheduler, a feature within the Oracle Database, to automate and manage the execution of various tasks or jobs at specified times

or under specific conditions.

Creating Jobs :-

- To schedule jobs with Oracle Scheduler, you first need to define the jobs you want to automate. Jobs can include tasks like running SQL scripts, PL/SQL procedures, or external scripts. These jobs are encapsulated as objects within the database.

Defining Programs :-

- For each job, you need to specify a program that defines what the job does. The program specifies the executable or procedure to run, as well as any required arguments or parameters. Programs can be created within the database or reference external executables.

Setting Schedules :-

- Once you have defined jobs and programs, you need to create schedules that determine when and how often these jobs should run. Schedules can be configured to execute jobs at specific times of the day, week, or month, and they can include options like specifying start and end times or recurring patterns.

Monitoring and Notifications :-

- Oracle Scheduler provides tools for monitoring the execution of scheduled jobs. You can view job status, logs, and performance metrics. Notifications can be configured to alert administrators when specific events or job outcomes occur.

Job History :-

- Oracle Scheduler maintains a history of job executions, allowing administrators to track the progress of jobs, view job status, and diagnose problems. This historical data can be valuable for auditing and troubleshooting purposes.

Security and Access Control :-

- Access control mechanisms ensure that only authorized users can create, modify, or execute scheduled jobs. This helps protect sensitive data and ensures that job scheduling is secure.

Job Management :-

- Administrators are responsible for creating, modifying, and deleting jobs. This includes defining the job type (e.g., SQL script, PL/SQL procedure), setting job attributes, specifying the program to execute, and assigning the job to schedules.

Programs and Executables :-

- Administrators manage programs and executables associated with jobs. This involves defining the programs, specifying executable files or PL/SQL procedures, and configuring program attributes.

Schedules :-

- Administering Oracle Scheduler involves the creation, modification, and deletion of schedules. Administrators determine when jobs should run by defining schedules, specifying start and end times, and configuring recurring patterns.

Dependency Management :-

- Administrators establish dependencies between jobs when needed. Dependencies define the order in which jobs are executed and ensure that specific conditions are met before a job starts.

Window Configuration :-

- Windows, which specify time intervals during which jobs are allowed to run, are set up and managed by administrators. Windows help control job execution during different time frames,

such as peak and off-peak hours.

Resource Allocation :-

- Resource plans and job classes are configured to manage resource allocation to jobs. Administrators ensure that critical jobs receive adequate system resources while preventing resource contention.

RDBMS Using ORACLE-2022

Q1-A

- 1.DBMS-Database Management System
- 2.TCL-Transaction Control Language
- 3.DML-Data Manipulation Language
- 4.DQL-Data Query Language

Q1-B

1. Formatting command SQL “COMPUTE SUM”

- The "COMPUTE SUM" clause in SQL is not a standard SQL command, but it might be specific to certain database systems or query tools.
- It's used for calculating the sum of values in a result set and displaying the total as a separate row in the query result.
- EX:-

```
SELECT product_id, sales_amount
FROM sales
COMPUTE SUM(sales_amount);
```

2. Formatting command SQL “BREAK ON”

- The "BREAK ON" command is not a standard SQL command, but rather a command that is specific to certain database systems, particularly Oracle SQL*Plus.
- It is used for formatting the output of SQL queries by controlling when breaks or summary lines should be displayed in the query result.
- EX:-

```
BREAK ON department_id SKIP 1
SELECT department_id, employee_id, salary
FROM employees
ORDER BY department_id, employee_id;
```

Q1-C

1. Characteristics of RDBMS

- RELATIONSHIP BETWEEN TWO TABLES OR FILE CAN BE SPECIFIED AT THE TIME OF TABLE CREATION.
- SUPPORTS CLIENT/SERVER ARCHITECTURE
- SUPPORTS DISTRIBUTED DATABASES
- MULTIPLE SECURITY LEVES; 1 LOGIN AT OS LEVEL 2 COMMAND LEVEL SECURITY 3 OBJECT LEVEL SECURITY
- MANY TABLES ARE GROUPED IN ONE DATABASE IN RDBMS

2. BCNF

- BCNF, or Boyce-Codd Normal Form, is a higher level of database normalization that is based on the functional dependencies of attributes within a relational database. BCNF is an important concept in database design and ensures that a database is structured in such a way that it avoids certain types of anomalies and redundancies.
- **Functional Dependency:** In a relational database, an attribute A is said to be functionally dependent on attribute B if, for every value of B, there is only one corresponding value of A. In other words, if you know the value of B, you can determine the value of A.
- BCNF, we need to satisfy the following two conditions:
- **The table must be in 1NF (First Normal Form):** This means that the table must have a primary key, and each attribute in the table must be atomic (indivisible).
- **For every non-trivial functional dependency $X \rightarrow Y$, X must be a superkey:** In BCNF, a non-trivial functional dependency means that Y is not a subset of X, and X is not a superkey. A superkey is a set of attributes that can be used to uniquely identify a tuple in a table.

Q1-D

1.Note On SQL *PLUS

- SQL stands for Structured Query Language.
- It is a domain-specific language used for managing and manipulating relational databases.
- SQL allows you to interact with databases to perform various tasks, including creating and modifying database structures, inserting, updating, and deleting data, and retrieving data for analysis and reporting.
- **Key Components of SQL:**
- SQL comprises several key components:
- **Data Definition Language (DDL):** DDL statements are used to define and manage the structure of a database. Common DDL statements include CREATE TABLE, ALTER TABLE, and DROP TABLE.
- **Data Manipulation Language (DML):** DML statements are used to interact with and manipulate data within a database. Common DML statements include SELECT, INSERT, UPDATE, and DELETE.

- **Data Query Language (DQL):** DQL is a subset of DML that specifically deals with querying data. The most common DQL statement is SELECT, which retrieves data from one or more tables.
- **Data Control Language (DCL):** DCL statements are used to control access and permissions within a database. Common DCL statements include GRANT and REVOKE.
- **Basic SQL Commands:**
 - SELECT: Used to retrieve data from one or more tables. It is often the most frequently used SQL command and supports various clauses such as WHERE, ORDER BY, and GROUP BY.
 - INSERT: Adds new rows of data to a table.
 - UPDATE: Modifies existing data in a table.
 - DELETE: Removes data from a table.
 - CREATE TABLE: Creates a new table with specified columns and data types.
 - ALTER TABLE: Modifies the structure of an existing table, such as adding, modifying, or deleting columns.
 - DROP TABLE: Deletes an entire table and its data.
- **Data Integrity:**
 - SQL databases enforce data integrity through various constraints, including:
 - Primary Key: Ensures that each row in a table is unique and has a unique identifier.
 - Foreign Key: Maintains referential integrity between tables by ensuring that values in one table correspond to values in another.
 - Unique Constraint: Enforces uniqueness of values within a column or a combination of columns.
 - Check Constraint: Allows you to define custom rules that data must adhere to.
- **SQL Database Management Systems (DBMS):**
 - SQL is used with various database management systems, such as MySQL, PostgreSQL, Oracle, Microsoft SQL Server, and SQLite. Each DBMS may have its own dialect of SQL, although many SQL statements are standardized.
- **Advanced SQL Topics:**
 - Joins: Combining data from multiple tables using JOIN operations (e.g., INNER JOIN, LEFT JOIN).
 - Subqueries: Embedding one SQL query within another.
 - Views: Virtual tables created from SQL queries, simplifying complex queries or providing a level of abstraction.

- Stored Procedures and Functions: Reusable code blocks written in SQL for data manipulation and automation.
- Transactions: A sequence of one or more SQL statements that are executed as a single unit of work, ensuring data consistency.

2. Dr. E.F. CODD RULES

- Rule 0 – Foundation rule
 - Any relational database management system that is propounded to be RDBMS or advocated to be a RDBMS should be able to manage the stored data in its entirety through its relational capabilities.
- Rule 1 - Rule of Information
 - Relational Databases should store the data in the form of relations. Tables are relations in Relational Database Management Systems.
- Rule 2 – Rule of Guaranteed Access
 - The use of pointers to access data logically is strictly forbidden. Every data entity which is atomic in nature should be accessed logically by using a right combination of the name of table, primary key represented by a specific row value and column name represented by attribute value.
- Rule 3 – Rule of Systematic Null Value Support
 - Null values are completely supported in relational databases. They should be uniformly considered as 'missing information'. Null values are independent of any data type.
- Rule 4 – Rule of Active and online relational Catalogue
 - In the Database Management Systems lexicon, 'metadata' is the data about the database or the data about the data. The active online catalogue that stores the metadata is called 'Data dictionary'.
- Rule 5 – Rule of Comprehensive Data Sub-language
 - A single robust language should be able to define integrity constraints, views, data manipulations, transactions and authorizations
- Rule 6 – Rule of Updating Views
 - Views should reflect the updates of their respective base tables and vice versa. A view is a logical table which shows restricted data. Views generally make the data readable but not modifiable. Views help in data abstraction.
- Rule 7 – Rule of Set level insertion, update and deletion
 - A single operation should be sufficient to retrieve, insert, update and delete the data.
- Rule 8 – Rule of Physical Data Independence

- Batch and end user operations are logically separated from physical storage and respective access methods.
- Rule 9 – Rule of Logical Data Independence
- Batch and end users can change the database schema without having to recreate it or recreate the applications built upon it.
- Rule 10 – Rule of Integrity Independence
- Integrity constraints should be available and stored as metadata in data dictionary and not in the application programs.
- Rule 11 – Rule of Distribution Independence
- The Data Manipulation Language of the relational system should not be concerned about the physical data storage and no alterations should be required if the physical data is centralized or distributed.
- Rule 12 – Rule of Non Subversion
- Any row should obey the security and integrity constraints imposed. No special privileges are applicable

Q2-A

- 1. Query within Query known as_____.
- Subquery
- 2. _____ command is used to insert data into the table.
- insert into tablename values('.....');
- 3. Join is command the data from two or more tables.(True/False).
- True
- 4. DELETE table command cannot be rollback(True/False).
- False

Q2-B

1. Explain COMMIT and ROLLBACK.

- COMMIT is a transaction control language in SQL. It lets a user permanently save all the changes made in the transaction of a database or table. Once you execute the COMMIT, the database cannot go back to its previous state in any case.
- When transaction begins its assigned to a rollback segment. Rollback segments store the “before” image of data which is ready to run in oracle .
- When creating image of data some time more memory spend to data which is useless so rollback segment undo these extra memory space and keep data in their perfect size.
- Oracle 9i allows you to undo to be managed in either of two ways:
- Manually
- Automatically

2. Explain GRANT and REVOKE commands.

- GRANT & REVOKE are the popular members of SQL family. these are the type of DCL commands that are used to assign permission to the users to perform a different task. The GRANT command is used for permitting the users whereas the REVOKE command is used for removing the authorization.
- When you create any object like table or etc you are owner of that object. So if you want other user use your object or they want to use your object you need to give them permission to access this is called granting of Privileges .
- Syntax of Granting permission using Grant statement :-
- Grant <object privileges>
- On <object name>
- To <UserName>
- [With Grant Option]
- Privileges once given can be taken back by the owner of the object This is call Revoking of Privileges.
- Syntax of Revoking permission :
- Revoke <object privileges>
- On <object Name>
- From <UserName>

Q2-C

1. Explain Having and Group by clause.

- The SQL HAVING Clause
- Having clause special use for give condition to group by like WHERE.
- You use the HAVING clause to filter groups of rows
- GROUP BY can be used without HAVING.
- But HAVING must be used with GROUP BY.
- You place the HAVING clause after your GROUP BY clause:
- HAVING Syntax
- SELECT column_name
FROM table_name
GROUP BY column_name
HAVING condition
ORDER BY column_name;
- The SQL GROUP BY Statement
- The GROUP BY statement groups rows that have the same values into summary rows, like "find the maxsal from each job in emp table".

- The GROUP BY statement is often used with aggregate functions (COUNT, MAX, MIN, SUM, AVG) to group the result-set by one or more columns.
- By default, GROUP BY sorts the rows in ascending order based on the values in the group column.
- GROUP BY Syntax
- SELECT column_name
FROM table_name
GROUP BY column_name
ORDER BY column_name;

2. Explain any three Aggregate Function.

- AVG : returns average value from table
- Select avg(sal) from emp
- MIN : returns minimum value from table
- Select min(sal) from emp
- MAX : returns maximum value from table
- Select max(sal) from emp
- SUM : return sum value
- Select sum(sal) from emp

Q2-D

1. Explain Data constraints in details.

- Constraints are set of rules used to enforce data integrity.
- They don't take space in database like data.
- Even if single column fails to satisfy constrain, entire record will be rejected by Oracle.
- **Types of constraints are :-**
 - Primary key
 - Foreign key
 - Unique
 - NOT NULL
 - Check
- **Feature of primary key:**
 - Does not allow duplicate values
 - NULL values not allowed
 - Primary key is not compulsory but recommended
 - Primary key can not be LONG or LONG RAW data type
 - Only one key is allowed per table
 - Unique Index created automatically on Primary key

- **---primary key at column level----**
- create table uni1
(ccode number(3) primary key,
corsename varchar2(15),
duration number(1));
- **---primary key at table level---**
- create table uni
(ccode number(3),
corsename varchar2(15),
duration number(1),
primary key(ccode,corsename));
- **Feature of foreign key:**
- It references a column of another table and it can be same table also
- Referenced field must be Unique or Primary key. The default is primary key
- Defined foreign key table is called child table where as defined primary key table is called parent table
- Child table may have duplicates and NULLS in foreign key column
- Parent record can be deleted only if child record does not exist
- Parent table can not be updated if child record exist
- **ON DELETE CASCADE :-**
- When the record is deleted in Parent table, all records in Child Table will be deleted automatically.
- **ON DELETE SET NULL :-**
- When record is deleted in Parent table, all field value held by foreign key of child table is set to NULL.
- Entire record will not be deleted.

- **Define foreign key:**
- create table college
(rno number(3),
sname char(20),
course_no number(3) references uni1(ccode));
on delete cascade
on delete set null

2. What is Join ? Explain different type of joins.

- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.

- **Different Types of SQL JOINS**

- Inner Join

- Outer Join

- Left outer Join

- Right outer Join

- Full outer Join

- Self Join

- Cross Join

- **Inner join:**

- The INNER JOIN keyword selects records that have matching values in both tables.

- This join is also known as equi join

- **Outer join:**

- SQL LEFT JOIN Keyword

- The LEFT JOIN keyword returns all records from the left table (table1), and the matched records from the right table (table2).

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- **Full outer join:**

- SQL FULL OUTER JOIN Keyword

- The FULL OUTER JOIN keyword return all records when there is a match in either left (table1) or right (table2) table records.

- **SQL Self JOIN:**

- A self JOIN is a regular join, but the table is joined with itself.

- **Cross join**

- In this join there is no condition given so records multiply by each other table

- Like if one table have 3 records and other table have 4 and we apply cross join than result will be 12 rows display

Q3-A

1. An Insert operation is not allowed when a view is created from multi table. (true/false) - True

2. List the types of snapshot.

Ans. Simple snapshot

Complex snapshot

3. The Oracle Engine automatically locks table data while executing SQL statements is called_____ locking.

Ans.DML

4. A NOWAIT option can be used to inform the oracle engine to terminate the sql statements if the record has already been locked. (true/false).

Ans.True

Q3-B

1. Describe DDL Lock.

- DDL locks (dictionary locks)
- DDL locks protect the structure of schema objects .
- It simply deals with descriptions of the database schema and is used to create and modify the structure of database objects in the database.
- DDL is a set of SQL commands used to create, modify, and delete database structures but not data.

2. Describe Latches.

- Protect internal database structures such as data files. Latches are visible only to the local instance in memory as opposed to locks which maintain information within the oracle database and are visible to all of the instances that have access to the database. As such, locks operate at the database level versus latches which are based at the instance level in oracle memory.

Q3-C

1. Explain lock Escalation.

- Lock escalation is a process that databases use to reduce the number of locks they hold.
- It occurs when a database raises the locks to a higher level on table.
- For example, if a single user locks many rows in a table, some databases automatically escalate the user's row locks to a single table.
- Locks in SQL Server can be quite expensive. For each lock the Lock Manager acquires, SQL Server must reserve memory – 64 bytes or 128 bytes. The amount depends on whether we are dealing with a 32-bit or 64-bit system, respectively.
- As the number of row locks on a table increases, SQL Server must acquire more and more memory. Hence, other processes are starving, out of memory.
- The table lock implies that no other session can modify its rows for the transaction duration, which may happen even when a blocking session does not manipulate all rows in the table.
- It is also worth mentioning that other factors can affect how locks are acquired and escalated in SQL Server. Those are the isolation level configured, indexing, and trace flags.

2. Explain Snapshot.

- Snapshot is a recent copy of the table from the database or a subset of rows/columns of a table.
- The SQL statement that creates and subsequently maintains a snapshot normally reads data from the database residing server.
- A snapshot is created on the destination system with the create snapshot SQL command. The remote table is immediately defined and populated from the master table.
- These are used to dynamically replicate data between distributed databases.
- **Two types of snapshots are available.**
Simple snapshots
Complex snapshots
- **Advantages :**
 - Response time is improved when local read-only copy of table exists.
 - Ease network loads.
 - Data subsetting.
 - Disconnected computing.
 - Mass deployment.
- **Disadvantages:**
 - Snapshots are not reachable when primary database goes offline.
 - It does not support full text indexing.
 - Snapshot runs out of disk if data changes frequently faster.
 - As no.of snapshots increases, disk space becomes problematic.

Q3-D

1. Write a note on bitmap index and application domain index.

- A bitmap is a mapping from one system such as integers to bits.
- It is also known as bitmap index or a bit array.
- The memory is divided into units for bitmap.
- These units may range from a few bytes to several kilobytes.
- Each memory unit is associated with a bit in the bitmap.
- If the unit is occupied, the bit is 1 and if it is empty, the bit is zero.
- The bitmap provides a relatively easy way to keep track of memory as the size of the bitmap is only dependent on the size of the memory and the size of the units.
- **Application Domain Index :**
 - Application domain indexes are custom indexes that are specific to the needs of your application.
 - They can be based on the semantics of your data and queries.

- **Example:**
- Suppose you have a table books with a publication_date column, and you want to create an index that takes into account the availability of books in a specific library branch.
- You could create an application domain index like this:
- CREATE INDEX idx_library_availability ON books (branch_id, publication_date);

Bitmap Indices Example

record number	ID	gender	income_level
0	76766	m	L1
1	22222	f	L2
2	12121	f	L1
3	15151	m	L4
4	58583	f	L3

Bitmaps for gender		Bitmaps for income_level	
m	10010	L1	10100
f	01101	L2	01000
		L3	00001
		L4	00010
		L5	00000

2. Write a note on pessimistic and optimistic locking.

- **Pessimistic**
- A Pessimistic approach is an approach of concurrency control algorithms in which the transaction is delayed if there is a conflict with each other at some point of time in the future.
- It locks the database's record for update access and other users can only access record as read-only or have to wait for a record to be 'unlocked'.
- Programming an app with a pessimistic concurrency approach can be more complicated and complex in managing because of deadlocks' risk.
- **Optimistic**
- An Optimistic approach is an approach of concurrency control algorithms that are based on assumption that conflicts of operations on a database are rare.
- It is advisable to run these transactions to completion and to check for conflicts only before they commit also here there's no checking to be done while the execution of transactions.
- This approach does not need any locking or time-stamping method. In an optimistic approach, a transaction is executed without any problems of restriction until transaction is committed.
- The optimistic approach allows the transactions to proceed in an unsynchronized way and also allows conflict checking at the end.
- This approach is also known as **validation** or **certification approach**.

Q4-A

- 1. Procedural/ structured query language
- 2. False

- 3. National Character Large Object
- 4. - Implicit cursor
- - Explicit cursor

Q4-B

1. Describe use of database trigger.

A database trigger is a predefined set of actions that automatically execute in response to specific events or changes in a database.

- These triggers are typically used to enforce data integrity, automate tasks, or log changes.
- **Syntax :-**
- CREATE OR REPLACE TRIGGER trigger_name
{BEFORE | AFTER | INSTEAD OF} {INSERT | UPDATE | DELETE}
ON table_name
[FOR EACH ROW]
DECLARE
 -- Declaration section
BEGIN
 -- Trigger code
EXCEPTION
 -- Exception handling, if needed
END;

2. Describe for loop in pl/SQL .

- In PL/SQL, a FOR loop is a control structure used to iterate over a range of values or through elements in a collection.
- **Example:-**
- BEGIN
 FOR i IN 1..5 LOOP
 DBMS_OUTPUT.PUT_LINE('Value of i: ' || i);
 END LOOP;
END;
- If you want the output in reverse order you can just add a keyword reverse before specifying the range.

Q4-C

1. Explain function.

- A function is a subprogram that is used to return a single value.
- You must declare and define a function before invoking it.

It can be declared and defined at a same time or can be declared first and defined later in the same block.

- **Syntax:**

- CREATE [OR REPLACE] FUNCTION function_name (parameters)
RETURN return_datatype IS/AS
[declaration_section]
BEGIN
[executable_section]
EXCEPTION
[exception_section]
END [function_name];

- We can also define some parameters while creating function.
- These parameters are:
 1. IN: It is a default parameter. It passes the value to the subprogram.
 2. OUT: It returns a value to the caller.
 3. IN OUT: It passes an initial value to the subprogram and returns an updated value to the caller.

- **Example:**

- create or replace procedure abc(a number, b number) as
add number(5);
function xyz(n1 number,n2 number) return number as
jadd number (10);
begin
jadd:=n1+n2;
return jadd;
end xyz;
begin
add:=xyz(a,b); --Function call
dbms_output.put_line('Add: ' || add);
end;

2. Explain exception handling.

- PL/SQL supports programmers to catch such conditions using EXCEPTION block in the program and an appropriate action is taken against the error condition.
- **There are two types of exceptions :**
 - 1 System-defined exceptions
 - 2 User-defined exceptions

1. system defined:-

No_data_found :-

- In This exception we need not declare exception, we can use directly in our exception part when in our pl/sql block no data match
- This exception generally use for SELECT statement, when no record match we can use this in our exception part

2. user defined:-

- To create user defined exception just follow this steps
- Declacre exception in Declaration part
Example : not_eligible exception;
- Raise Declared exception in Execution part
Example : raise not_eligible;
- In last our Exception part just put your declared exception in between WHEN and THEN for proper message to our exception
Example : when not_eligible then

Q4-D**1. Write a note on objects.**

- In Oracle, "objects" typically refer to database objects.
- They are the entities used to store, manage, and manipulate data within an Oracle database.
- Some common types of database objects in Oracle include:
 1. Table
 2. Views
 3. Sequence
 4. Indexes
 5. Synonyms
- **1. Tables:**
 - Tables are used to store data in rows and columns.
 - They are the fundamental structure for organizing and storing data in a relational database.
 - Example:


```
create table stud
( no number(2),
  name char(20));
```
- **2. Views:**
 - An Oracle view is a virtual table that physically doesn't exist.

- It stores only definition in oracle data dictionary in the form of SQL statement
- Creating view does not take any storage space.
- A view is created by a query joining one or more tables.

- **Example:**

```
create view v1
as select ename, sal,deptno
from emp
where deptno=20;
```

- **3. Sequences:**

- Sequence produces a series of numerical values in accordance with the specification used to create it.
- It automatically generates the primary key and unique key values.
- It may be in ascending or descending order and can be used in multiple tables.

- **Example :**

- create sequence seq1;

- **4.Indexes:**

- Indexes are data structures that improve the retrieval speed of rows from a table.
- They are used to speed up SELECT statements by providing fast access to data based on certain columns.

- **5.Synonyms:**

- Synonyms are aliases for database objects.
- They simplify access to objects and provide a level of abstraction.

2. Write a note on nested table.

- Nested tables are similar to PL/SQL tables, but they can
- be stored in the database as a column in a table or as
- a database object.
- Nested tables are single-dimensional, unbounded collections of homogeneous elements.
- First, a nested table is single-dimensional, meaning that each row has a single column of data like a one-dimension array.
- Second, a nested table is unbounded. It means that the number of elements of a nested table is predetermined.
- Third, homogeneous elements mean that all elements of a nested table have the same data type.

- **example to declare a nested table:-**

- CREATE TABLE Orders
(order_id NUMBER PRIMARY KEY,
products NestedTableType)
NESTED TABLE products STORE AS ProductsTable;

- **example:-**

- -- creation of object:
- create type obj as object
(m1 number(3),m2 number(3),m3 number(3),m4 number(3));
- -- creation of nested table of obj type:
- create type tnest as table of obj;
- -- creation of nested table:
- create table stud_tab
(sname char(30),
smark tnest)
nested table smark store as tnest_tab;
- --Inserting in nested table:
- insert into stud_tab values ('&sname',tnest(obj(&m1,&m2,&m3,&m4)));

Q5-A

- 1. - Backup and Recovery
 - - Space Management
 - - Patching and Upgrades
 - - Security Updates
- 2. True
- 3. SMON- System Monitor Process
- 4. CKPT- Checkpoint process

Q5-B

1. Describe oracle export utility.

- Oracle provides an export utility called "exp" (or "expdp" for Data Pump) allows you to extract data, schemas, tables, and other database objects into a binary export file.
- You can use it to generate a dump file containing data and metadata.
- There are two main versions of the Oracle Export Utility:
 1. Traditional Export (exp)
 2. Data Pump Export (expdp)

2. Describe control files .

- The control files of database is the binary file that contains a great deal of database information.
- The control files contain name , data about database log file .
- Oracle allows you to maintain duplicate copies with the help of control files.

Q5-C

1. Explain table space.

- A table space is a storage structure containing tables, indexes, large objects, and long data.
- They are used to organize data in a database into logical storage groupings that relate to where data is stored on a system.
- Table spaces are stored in database partition groups.
- Table space is logical storage unit within as oracle database.
- Its logical Because a tablespace is not visible in the file system of the machine on which The database stores
- **Create tablespace :**
- Create tablespace <tbasp_name> ' <datafile_path>' Size<n M/K/G>
EXTENT ANAGEMENT LOCAL AUTOLOCATTE;
- **Alter tablespace :**
- ALTER TABLESPACE TABS1 ADD DATAFILE 'PATH' SIZE <N M/K/G>;
- **Drop tablespace :**
- DROP TABLESPACE TABSP1 INCLUDING CONTENTS;

2. Explain initialization parameters

- The initialization parameter file is a text file that contains a list of parameters and a value for each parameter.
- The file should be written in the client's default character set. Specify values in the parameter file which reflect your installation.
- The following are sample entries in a parameter file:
 PROCESSES = 100
 OPEN_LINKS = 12
 GLOBAL_NAMES = TRUE

- **Database administrators can use initialization parameters to do the following:**
- optimize performance by adjusting memory structures, for example, the number of database buffers in memory
- set some database-wide defaults, for example, how much space is initially allocated for a context area when it is created
- set database limits, for example, the maximum number of database users
- specify names of files
- **The following rules govern the specification of parameters in the parameter file:**
- All parameters are optional.
- Only parameters and comments should appear in the parameter file.
- A pound sign (#) starts a comment; the rest of the line is ignored.
- The server has a default value for each parameter. This value may be operating system dependent, depending on the parameter.
- Parameters can be specified in any order.
- Case (upper or lower) in filenames is only significant if case is significant on the host operating system.

Q5-D

1. Write a note on rollback segment.

- In the ORACLE database, the rollback segment is a structure that stores previous data during transactions.
- To support the rollback facility we have a rollback segment in the ORACLE database.
- If during a transaction any failures occur then it Makes sure that changes are not reflected back Into our original Database.
- It helps us in restoring our database in Consistent state in case of any failure. To ensure Data consistency and isolation property.
- The rollback segment is used to recover our data In case of database failure.
- With the help of the Rollback command, we can Restore our previous data which is stored inside The rollback segment to our original database.
- When transaction begins, its assigned to a rollback segment. Rollback segments store the “before” image of data which is Ready to run in oracle.
- When creating image of data some time more memory spend to Data which is useless so rollback segment undo these extra Memory space and keep data in their perfect size.

- **Oracle 9i and above version allows you to undo space to be managed in either of two ways :-**
 1. Manually
 2. Automatically
- **Manually :-**
 - This mode is commonly referred to as rollback managed undo (RMU).
 - Its invoke by setting the parameter UNDO_MANAGEMENT to MANUAL
- **Automatically :-**
 - This mode is commonly referred to as automatically managed undo (AMU). Its invoke by setting the parameter UNDO_MANAGEMENT to AUTO.
- **Create rollback segment :-**
 - CREATE [PUBLIC] ROLLBACK SEGMENT rbs_name option(s)

2. Write a note on schedule architecture.

- Oracle Database includes Oracle Scheduler, an enterprise job scheduler to help you simplify the scheduling of hundreds or even thousands of tasks.
- Oracle Scheduler (the Scheduler) is implemented by the procedures and functions in the DBMS_SCHEDULER PL/SQL package.
- **Job:** A job is a specific task or unit of work that you want to automate using Oracle Scheduler.
- **Program:** A program defines what a job does.. A job references a program to execute a specific task.
- **Schedule:** A schedule defines when and how often a job should run. It specifies the frequency, start time, end time, and recurrence pattern for executing a job.
- **Job Class:** Job classes are used to categorize and manage jobs. You can assign jobs to specific job classes based on their characteristics or priorities.
- **Window:** A window is a specific time interval during which jobs are allowed to run.
- **Dependency:** Dependency defines the relationship between jobs. You can specify that one job should only run after another job has completed successfully.
- "Scheduling Jobs with Oracle Scheduler" involves using Oracle Scheduler, a feature within the Oracle Database, to automate and manage the execution of various tasks or jobs at specified times or under specific conditions.

- **Creating Jobs:** Jobs can include tasks like running SQL scripts, PL/SQL procedures, or external scripts. These jobs are encapsulated as objects within the database.
- **Defining Programs:** For each job, you need to specify a program that defines what the job does. Programs can be created within the database or reference external executables.
- **Setting Schedules:** Once you have defined jobs and programs, you need to create schedules that determine when and how often these jobs should run.
- **Monitoring and Notifications:** Oracle Scheduler provides tools for monitoring the execution of scheduled jobs. You can view job status, logs, and performance metrics.
- **Job History:** Oracle Scheduler maintains a history of job executions, allowing administrators to track the progress of jobs, view job status, and diagnose problems.
- **Security and Access Control:** Access control mechanisms ensure that only authorized users can create, modify, or execute scheduled jobs.